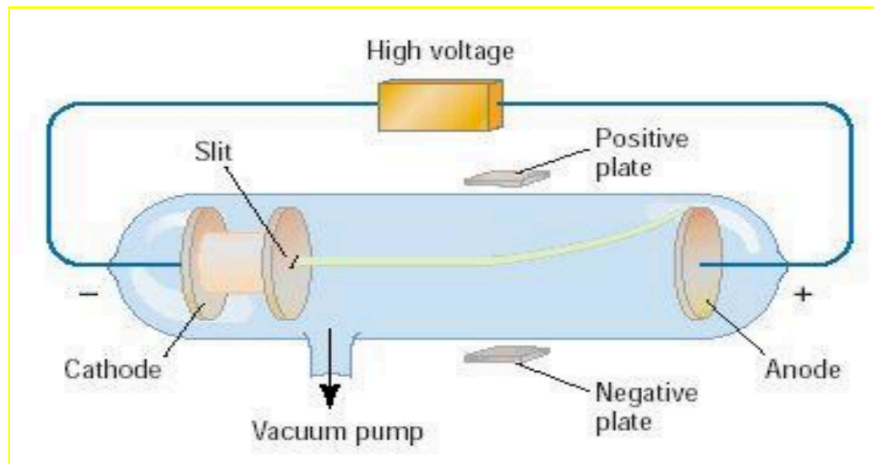


Atomic Structure

Discovery of Electron

Discovery of the Electron

In 1897, J.J. Thomson used a *cathode ray tube* to deduce the presence of a negatively charged particle: the **electron**



Conclusions from the Study of the Electron

- Cathode rays have identical properties regardless of the element used to produce them.
- All elements must contain identically charged electrons.
- Atoms are neutral, so there must be positive particles in the atom to balance the negative charge of the electrons.
- Electrons have so little mass that atoms must contain other particles that account for most of the mass.

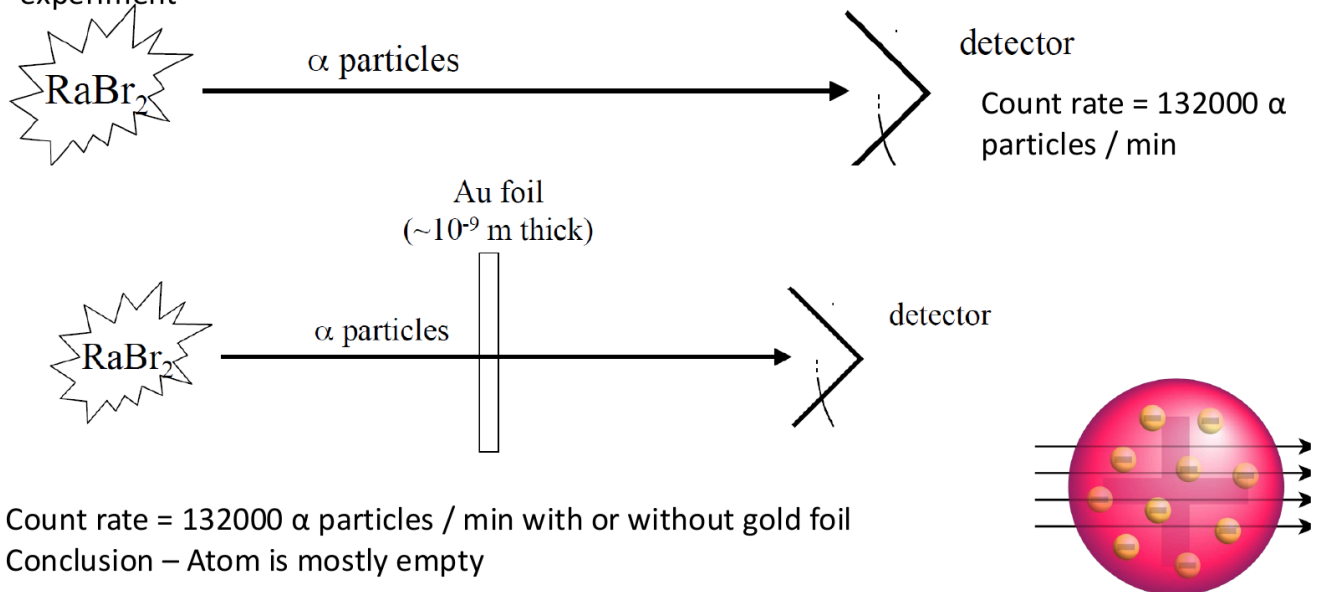
Thomson's Plum Pudding Model

Negative charges embedded within larger spheres of positive charge.

Discovery of the Nucleus

- Ernest Rutherford (Cambridge University, England, 1871-1937) studied α emission from newly-discovered radioactive elements that he got from his friend Marie Curie.
- Radium Bromide Shipped to Rutherford for experiment from France to England

With postdocs Hans Geiger and undergraduate E. Marsden, the Rutherford Lab did the experiment



The Experiment

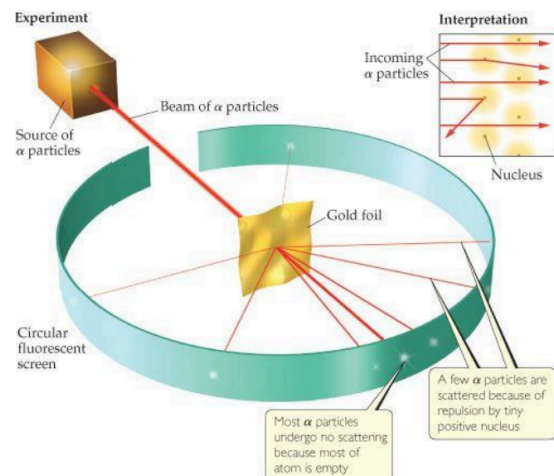
Little advance experiment from the Rutherford Laboratory: Now the detector is all around for detection of back scatter alpha particles

Mass of alpha particle: 8000 electron mass
Speed: 2×10^7 m/s

Count rate on backscatter side = 20 α particles / min

Probability (P) of backscattering:

$$P = \frac{\text{count rate backscattered}}{\text{count rate of incident particles}} = \frac{20}{132000} = 2 \times 10^{-4} \text{ (0.02\%)}$$



Conclusion

- The gold atoms are mostly empty.
- A very tiny percentage of α particles hit something massive in the atom and backscatter (bounce back). This observation indicates that most of the mass of the atom is concentrated in a very small volume relative to the volume of the entire atom. We now call this the NUCLEUS.
- Rutherford proposed the charge on the nucleus to be positive, since electrons are negatively charged and atoms are neutral. Charge of the electrons in an atom = $-Ze$. Z = atomic number e = absolute value of electron's charge. Charge of the nucleus of an atom = positive = $+Ze$

- Rutherford calculated the diameter of the nucleus to be $10^{-14}m$. His calculation related the probability of backscattering to the diameter of the nucleus based on the size of the atom and the thickness of the foil.

Rutherford Scattering formula (Annexure of Chapter 4 – PLEASE DO. It will be taught in detail by Prof. Shandilya)

The formula that Rutherford obtained for alpha particle scattering by a thin foil on the basis of the nuclear model of the atom is

Rutherford
Scattering Formula

$$N(\theta) = \frac{N_i n t Z^2 e^4}{(8\pi\epsilon_0)^2 r^2 K E^2 \sin^4(\frac{\theta}{2})}$$

$N(\theta)$ = number of alpha particles per unit area that reach the screen at a scattering angle of θ

N_i = total number of alpha particle that reach the screen

n = number of atoms per unit volume in the foil

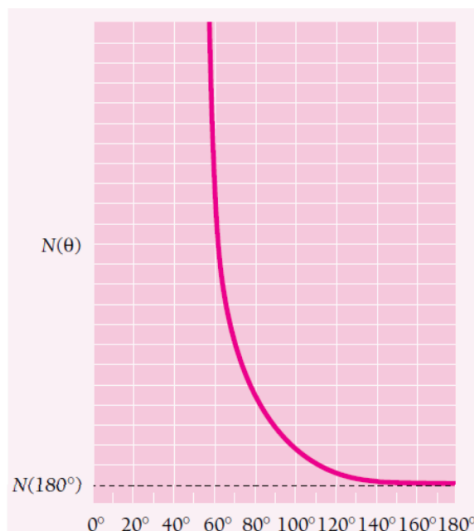
Z = atomic number of the foil atoms

r = distance of the screen from the foil

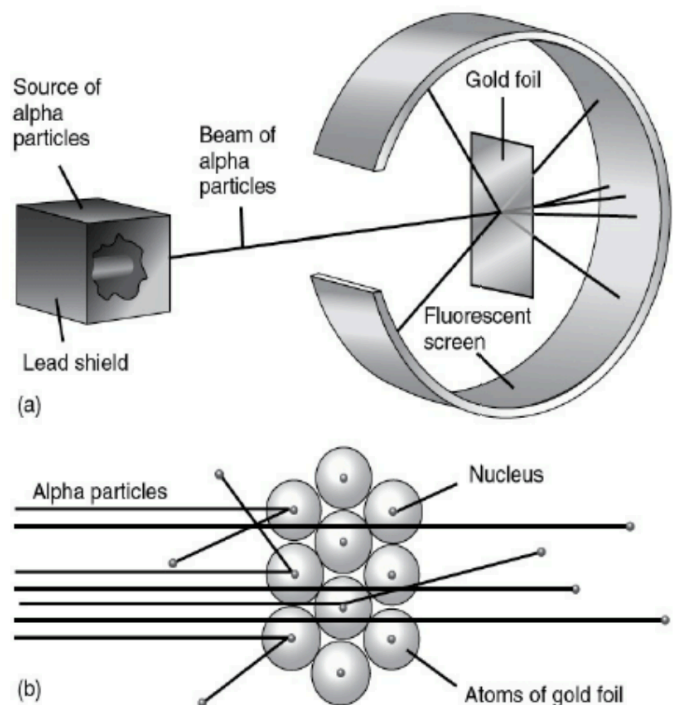
KE = kinetic energy of the alpha particles

t = foil thickness

The predictions of Scattering Formula agreed with the measurements of Geiger and Marsden, which supported the hypothesis of the nuclear atom.



Because $N(\theta)$ is inversely proportional to $\sin^4(\theta/2)$ the variation of $N(\theta)$ with θ is very pronounced : only 0.14 percent of the incident alpha particles are scattered.



Nuclear Dimensions

Rutherford assumed $\rightarrow$ the size of a target nucleus $<$ the minimum distance R (the distance the incident alpha particles approach the nucleus before being deflected away)

R = The distance of **closest approach** R for the most energetic alpha particles.

Rutherford scattering $\rightarrow$ gives us a way to find an upper limit to nuclear dimensions.

An alpha particle will have its **smallest R** when **it approaches a nucleus head on**, which will be followed by a **180° scattering**.

At the instant of closest approach the initial kinetic energy (KE) of the particle is entirely converted to electric potential energy, and so at that instant

$$KE_{initial} = PE = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{R}$$

since the charge of the alpha particle is $2e$ and that of the nucleus is Ze .

Distance of closest approach

$$R = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{KE_{initial}}$$

The maximum KE found in alpha particles of natural origin is 7.7 MeV,

$$7.7 \text{ MeV} = 1.2 \times 10^{-12} \text{ J}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\begin{aligned} R &= \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{KE_{initial}} \\ &= \frac{9 \times 10^9 \text{ Nm}^2 (2Z)(1.6 \times 10^{-19} \text{ C})^2}{\text{C}^2 \frac{1.2 \times 10^{-12} \text{ J}}{}} \\ &= 3.8 \times 10^{-16} Z \text{ m} \end{aligned}$$

The atomic number of gold is $Z=79$

$$R = 3.0 \times 10^{-14} \text{ m}$$

Radius of the gold nucleus $< 3.0 \times 10^{-14} \text{ m} \rightarrow$ less than 10^{-4} times of the atomic radius

Electron Orbits

ELECTRON ORBITS

Hydrogen atom : Single electron.

Assume a circular electron orbit for convenience.

The centripetal force

$$F_c = \frac{mv^2}{r}$$

holding the electron in an orbit r from the nucleus is provided by the electric force between them

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

The condition for a dynamically stable orbit is

$$F_c = F_e$$

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$\text{Electron velocity, } v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$$

The total energy E of the electron in a hydrogen atom is the sum of its kinetic and potential energies, which are

$$KE = \frac{1}{2}mv^2 \quad PE = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

The minus sign $\rightarrow$ $PE = 0$ at $r = \infty$, that is, when the electron and proton are infinitely far apart

$$E = KE + PE = \frac{1}{2}mv^2 - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$$E = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r} - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

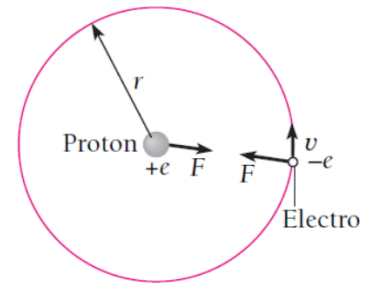
Total energy of hydrogen atom

$$E = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{r}$$

The total energy of the electron is negative.

This holds for every atomic electron and reflects the fact that **it is bound to the nucleus.**

If $E > \text{zero} \rightarrow$ an electron would **not follow a closed orbit around the nucleus.**



(FAILURE OF) THE CLASSICAL DESCRIPTION OF AN ATOM

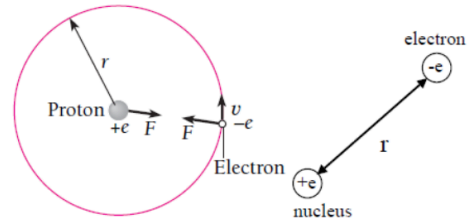
How is an electron bound to the nucleus? What is the force between them? **Coulomb's Force Law** describes the force of attraction and repulsion between charged particles.

$$\text{Coulomb Force } (F_r) = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$$

Q_x = charge on particle x

r = distance between two charges

ϵ_0 = permittivity constant of a vac
($8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$)



When charges are the same sign, acceleration of the particles will push them apart. The force is positive and repulsive.

When charges are the opposite sign, acceleration will pull them together. The force is negative and attractive.

Consider an H atom ($Z=1$) with 1 electron and 1 proton.
When $r \rightarrow \infty$, $F(r) = 0$

If the electron and the nucleus are infinitely attracted, why do they stay apart.

As $r \rightarrow 0$, $F(r) = \infty \rightarrow$ infinitely attractive

Coulomb's law only describes force as a function of r (distance). It doesn't tell us how r changes with time, t .

Newton's Second law:

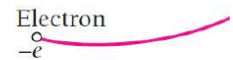
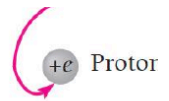
Force, $F = ma$ (mass X acceleration)

$F = m (dv/dt)$ (function of velocity) = $m(d^2r/dt^2)$ (function of distance)

Plug in Coulomb's force law

For $r_{\text{initial}} = 0.5 \times 10^{-10} \text{ m}$, calculate how fast it should go to zero. The answer is very fast.

$r=0$ at $t = 10^{-10} \text{ s}$



This predicts that the electron should crash into the nucleus in 10^{-10} sec. But why is it not doing that?

The laws of classical Mechanics no longer work at this size scale.

Atomic Spectra and Bohr's Model

- Bohr's postulates were based on the observation of discrete spectra and the results of the α -scattering experiment of Rutherford.
- Electrons move around the nucleus in discrete orbits.
- These orbits correspond to fixed energies and are stable (i.e. no radiation is emitted by such electrons)
- Radiation is emitted only when the electron jumps from one discrete orbit to another orbit of a lower energy. When electrons absorb radiation, the reverse transition occurs.
- A dynamic equilibrium exists between the centrifugal force and the coulomb force between the electron and the nucleus.

$$\frac{(Ze)^2}{4\pi\epsilon_0 r^2} = m\omega^2 r \rightarrow \omega = \frac{e^2}{4\pi\epsilon_0 m r^3}$$

The electron revolves at the frequency ω in the orbit. Classical electrodynamics predicts that the charge will radiate at this very frequency and lose energy. Bohr's postulates override this.

The total energy of the electron is

$$\frac{(Ze)^2}{4\pi\epsilon_0 r^2} = m\omega^2 r \rightarrow \omega = \frac{e^2}{4\pi\epsilon_0 m r^3}$$

$$E = E_{kin} + E_{pot} = \frac{1}{2} m \omega^2 r^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

Using the above obtained value of ω , we get

$$E = -\frac{e^2}{8\pi\epsilon_0 r} = \frac{1}{2} \frac{\omega^{2/3}}{(4\pi\epsilon_0)^{2/3}} (me^4)^{1/3}$$

Bohr's important insight was to recognize that the frequency of radiation (according to the postulates) should be given by

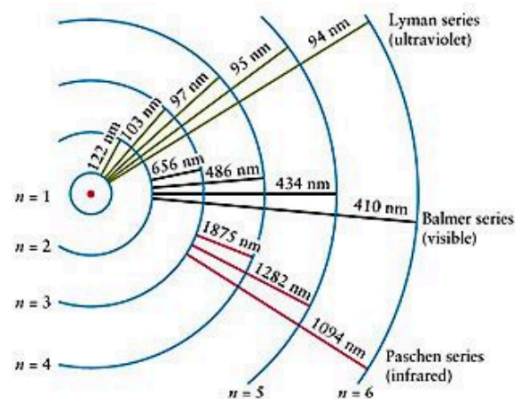
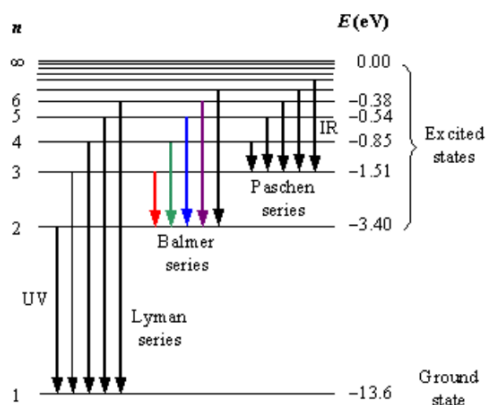
$$h\nu = E_n - E_{n'}$$

where $h\nu$ is the radiation quantum as proposed by Planck and that this radiation must be the same as that.

Balmer series - wavelengths in the visible portion of the hydrogen spectrum.

Lyman series - wavelengths in the ultraviolet region > smallest is 91.2 nm > 13.598eV

Rydberg constant = 10^7 m^{-1} $1.097 \times 10^7 \text{ m}^{-1}$



Hydrogen Series of Spectral Lines $\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$

Discoverer (year)	Wavelength	n	k
Lyman (1916)	Ultraviolet	1	>1
Balmer (1885)	Visible, ultraviolet	2	>2
Paschen (1908)	Near Infrared	3	>3
Brackett (1922)	Infrared	4	>4
Pfund (1924)	Far Infrared	5	>5

Rydberg-Ritz Combination Principle

The wavenumber of any spectral line in an atom's spectrum can be expressed as the sum or difference of the wavenumbers of two other spectral lines from the same element

THE BOHR ATOM

The de Broglie wavelength of an electron is $\lambda = \frac{h}{mv}$

v = electron velocity

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$$

Orbital electron wavelength

$$\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$$

Radius of electron orbit, $r = 5.3 \times 10^{-11} \text{ m}$
 The electron wavelength $\lambda = \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{1.6 \times 10^{-19} \text{ C}} \sqrt{\frac{(4\pi)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(5.3 \times 10^{-11} \text{ m})}{9.1 \times 10^{-31} \text{ kg}}}$
 $= 33 \times 10^{-11} \text{ m}$

This wavelength is exactly the same as the circumference of the electron orbit,

$$2\pi r = 33 \times 10^{-11} \text{ m}$$

Only waves with wavelengths that just fit in survive (all others cancel themselves out)

Bohr's Orbit: An electron can circle a nucleus only if its orbit contains an integral number of de Broglie wavelengths.

Condition for orbit stability

$$n\lambda = 2\pi r_n \quad n = 1, 2, 3 \dots$$

r_n = the radius of the orbit that contain n wavelengths.

n = the **quantum number** of the orbit.

Substituting for λ , the electron wavelength

$$\frac{nh}{e} \sqrt{\frac{4\pi\epsilon_0 r_n}{m}} = 2\pi r_n$$

Orbital electron wavelength $\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$

The possible electron orbits are those whose radii



$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} \quad n = 1, 2, 3, \dots$$

The innermost radius is called Bohr Radius

Bohr radius

$$a_0 = r_1 = 5.292 \times 10^{-11} \text{ m}$$

The other radii in terms of Bohr Radius are

$$r_n = n^2 a_0$$

ENERGY LEVELS AND SPECTRA

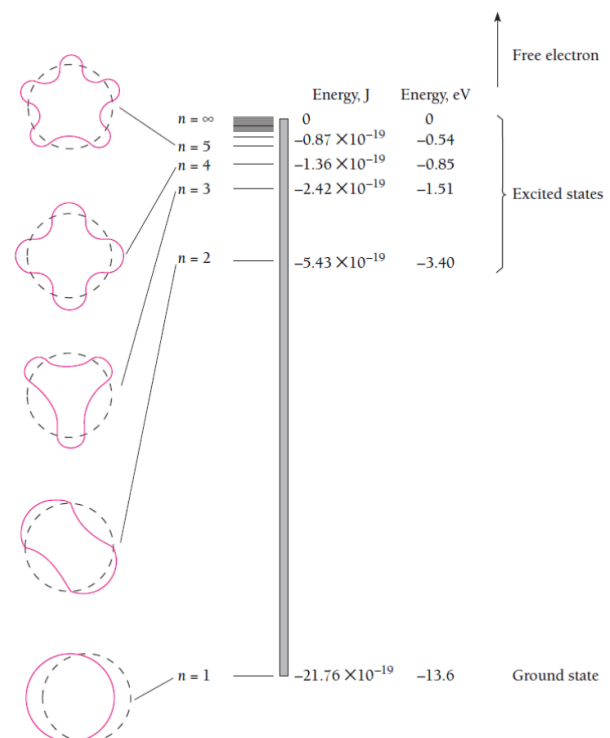
The electron energy E_n is given in terms of the orbit radius r_n as

$$E_n = -\frac{e^2}{8\pi\epsilon_0 r_n}$$

The **energy levels** of the hydrogen atom

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \frac{E_1}{n^2} \quad n = 1, 2, 3, \dots$$

$$E_1 = -2.18 \times 10^{-18} \text{ J} = -13.6 \text{ eV}$$



Bohr's Correspondence Principal

approaches classical mechanics.

In limit that $n \rightarrow \infty$,

quantum mechanics must agree with classical physics

$$E_{\text{photon}} = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = hf_{\text{photon}}$$

$$n_i \approx n_f \text{ when both } n_i, n_f \gg 1$$

Bohr's correspondence principle considers transitions between **nearby very-high- n states**, for example

$$n_i = n, n_f = n - 1,$$

with

$$n \rightarrow \infty.$$

Although n and $n - 1$ always differ by exactly 1, their **relative difference**

$$\frac{n_i - n_f}{n_i} = \frac{1}{n}$$

goes to zero as $n \rightarrow \infty$.

For very large n , we can approximate

$$\frac{1}{(n-1)^2} \simeq \frac{1}{n^2} \left(1 + \frac{2}{n} \right),$$

so

$$\frac{1}{(n-1)^2} - \frac{1}{n^2} \simeq \frac{2}{n^3}.$$

Therefore,

$$hf \simeq \frac{27.2 \text{ eV}}{n^3}.$$

The beautiful part of the correspondence principle $\rightarrow$ If you calculate the **classical orbital frequency** of the electron in the n -th Bohr orbit,

$$f_{\text{orbit}} = \frac{v_n}{2\pi r_n},$$

v_n = speed of the electron in the n -th Bohr orbit.

you obtain exactly the same large- n result as the photon frequency:

$$f_{\text{photon}} \rightarrow f_{\text{orbit}}$$

for

$$n \rightarrow \infty, \Delta n = 1.$$

For very large quantum numbers, transitions between neighbouring quantum states produce radiation whose frequency approaches the classical orbital frequency of the electron.

This is Bohr's correspondence principle:

Quantum physics $\rightarrow$ classical physics in the appropriate large-quantum-number limit.

$n = 1, 2, 3$	:	levels clearly discrete
$n = 100, 101, 102$	:	levels extremely close
$\downarrow$		
$n \rightarrow \infty$	:	almost a continuum $\rightarrow$ classical behaviour

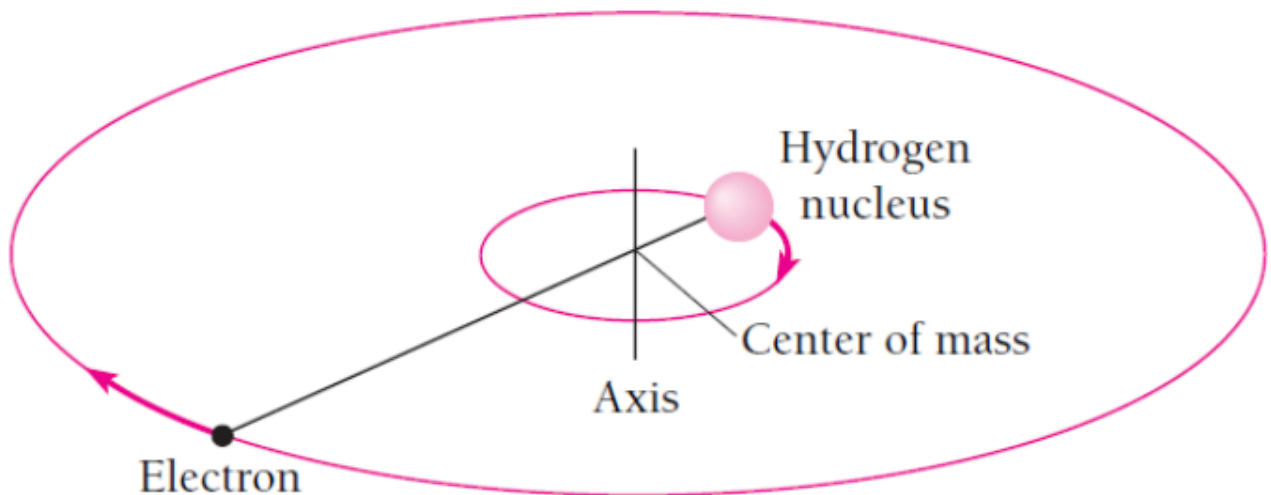
Conclusion: Spectroscopy provides direct evidence for **quantization of energies** (electronic, vibrational, rotational etc.) of atoms and molecules

Limitations of Bohr Model & Rydberg-Ritz Equation

- The model only works for hydrogen (and other one electron ions) – **ignores e-e repulsion**.
- Does not explain **fine structure** of spectral lines.
- Note:** The Bohr model (assuming circular electron orbits) is fundamentally incorrect.

Nuclear Motion

in reality, both the nucleus and electron orbit their common center of mass, which lies near the nucleus due to its much greater mass. This system is equivalent to a single particle of mass m' orbiting the heavier nucleus.



Reduced mass

$$m' = \frac{mM}{m + M}$$

m = electron mass
 M = nuclear mass

To account for the nucleus's motion in a hydrogen atom, we replace the electron with a particle of mass m' . The atom's energy levels then become

Energy levels corrected for nuclear motion

$$E'_n = -\frac{m'e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \left(\frac{m'}{m}\right) \frac{E_1}{n^2}$$

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \frac{E_1}{n^2}$$

$$-\frac{e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \frac{E_1}{mn^2}$$

$$E_1 = -\frac{e^4}{8\epsilon_0^2 h^2}$$

Due to the nucleus's motion, all hydrogen energy levels shift by the fraction

$$\frac{m'}{m} = \frac{M}{M + m} = 0.99945$$

This represents an increase of 0.055 percent because the energies E_n , being smaller in absolute value, are therefore less negative.

Energy levels
corrected for
nuclear motion

$$E'_n = -\frac{m'e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \left(\frac{m'}{m} \right) \left(\frac{E_1}{n^2} \right)$$

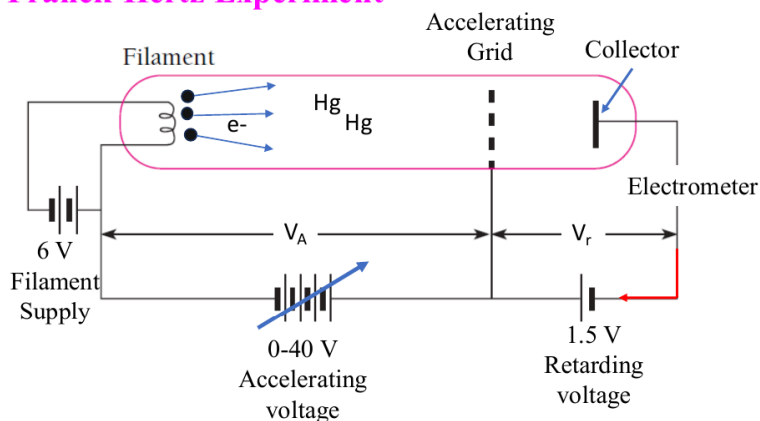
H atom $E = -13.606 \text{ eV}$ $\rightarrow$ -13.598 eV $\leftarrow$ reduced mass

Experiment $E = -13.598 \text{ eV}$

- The energy equation removes the discrepancy between the predicted wavelengths of the spectral lines of hydrogen and the measured ones.
- The value of the **Rydberg constant R** without correcting for nuclear motion is $1.0973731 \times 10^7 \text{ m}^{-1}$;
- The correction lowers **R** to $1.0967758 \times 10^7 \text{ m}^{-1}$.
- The notion of reduced mass played an important part in the discovery of deuterium,
- Deuterium nucleus contains a neutron as well as a proton.
- Because of the greater nuclear mass, the spectral lines of deuterium are all shifted slightly to wavelengths shorter than the corresponding ones of ordinary hydrogen.
- Ha line of deuterium > Transition $n = 3 > n = 2$ energy level > Wavelength 656.1 nm,
- Ha line of Hydrogen > Transition $n = 3 > n = 2$ energy level > Wavelength 656.3 nm.
- Identification of deuterium in 1932 by the American chemist Harold Urey.

Frank-Hertz Experiment

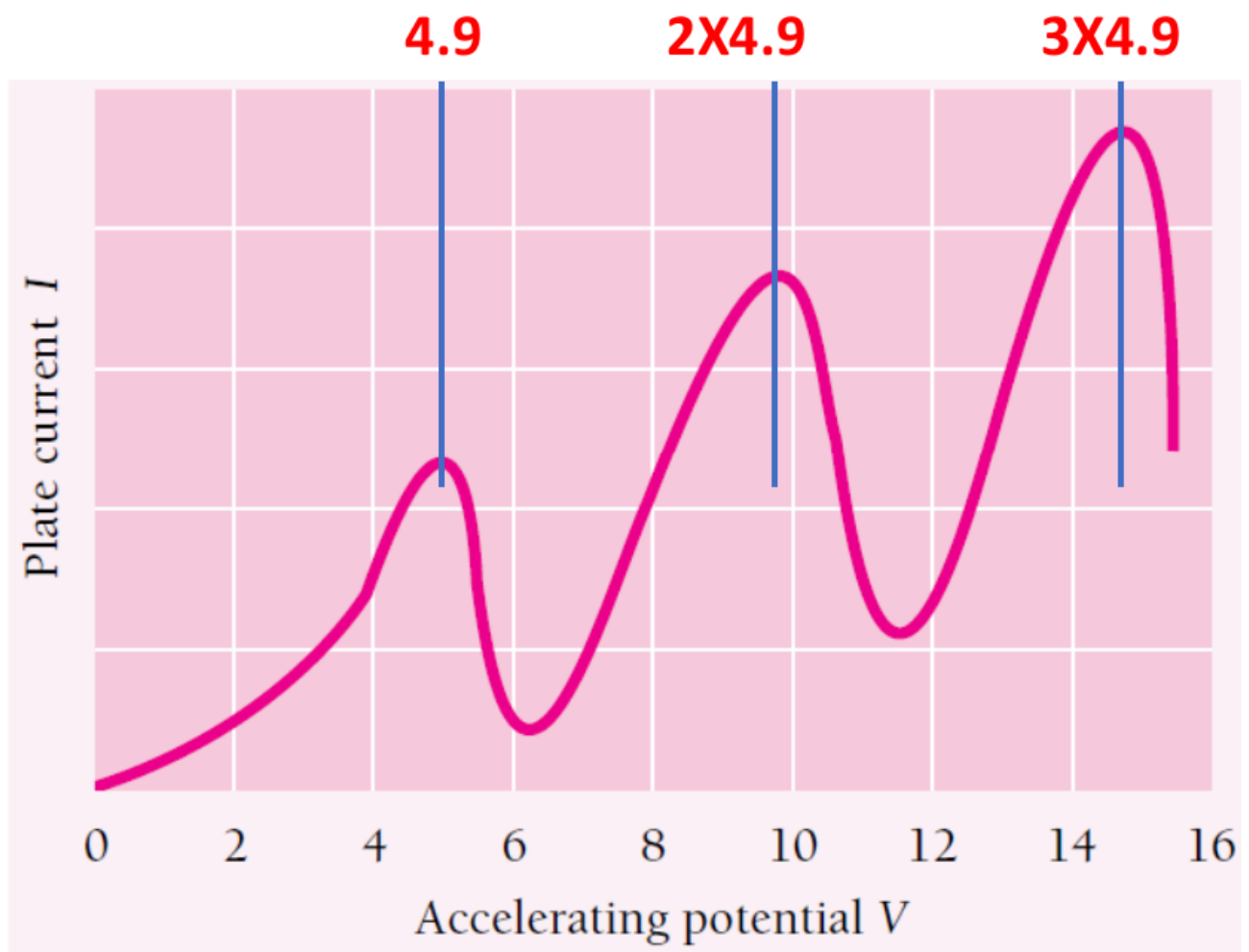
Franck-Hertz Experiment



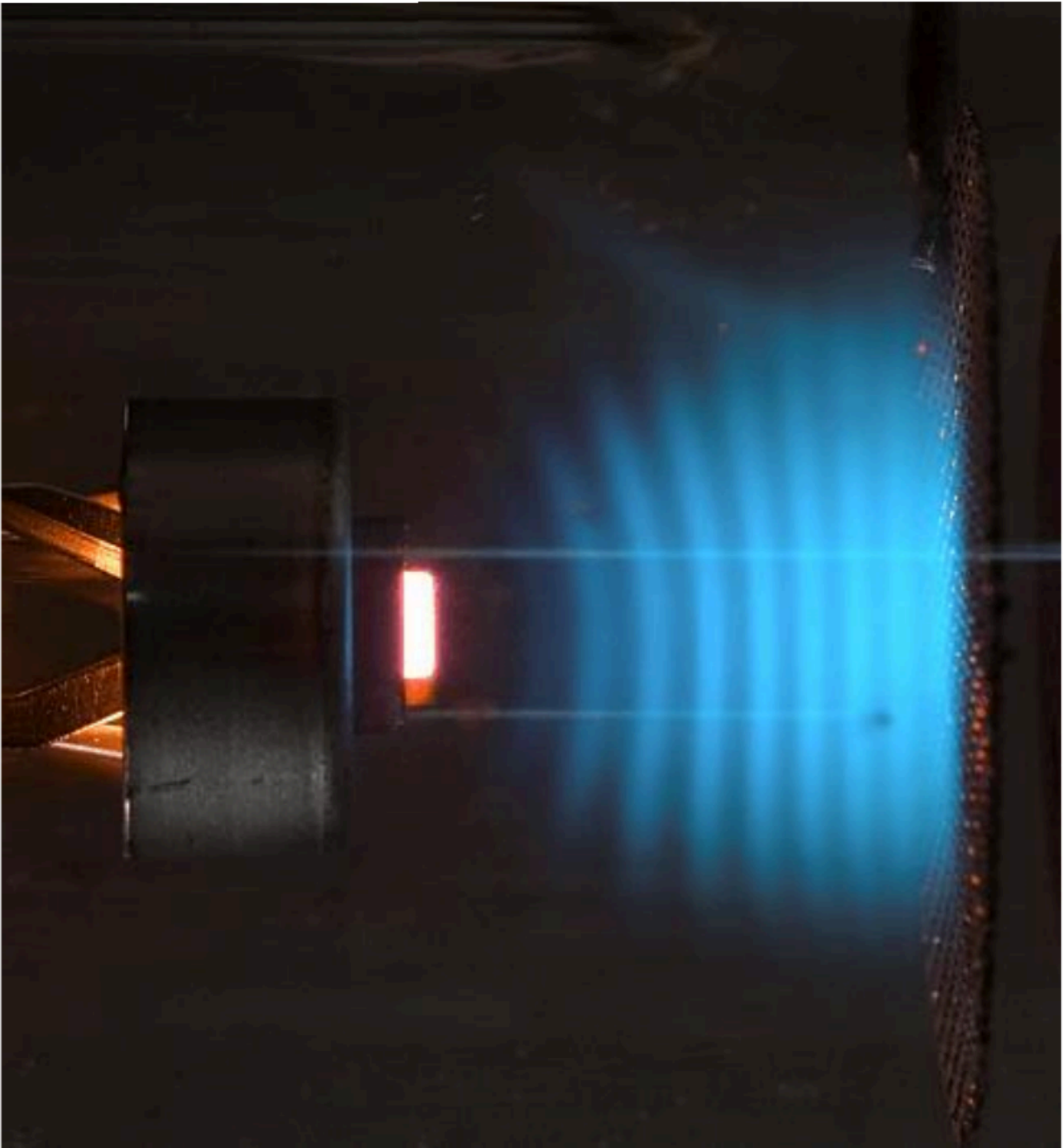
- The Filament ejects electrons in the gas filled tube.
- A variable (positive) accelerating voltage, V_A , is applied to the grid.
- The electron acquires kinetic energy $K = eV_A$

- There is (negative) retarding voltage (V_r) between the grid and collector.
- Only electrons having enough energy will overcome this potential and reach the collector and contribute to current measured with an Electrometer.

- If $V_r > V_A \rightarrow$ no electron can reach the collector, so no current would be measured.
- If $V_r < V_A \rightarrow$ if the tube is highly evacuated, most of the electrons would reach the collector and have energy $|e|(V_A - V_r)$
- Franck and Hertz observed the collector current as a function of $V_A (> V_r)$ when tube was filled with various gases (result for mercury gas is shown here)
- At first, the current increased as was expected for a typical vacuum tubes, **but as $\sim 4.9\text{V}$ current suddenly dropped.**
- Then, the increase resumed until 9.8 V , and so on.



- The current drops because fewer electrons reach the collector.
- This occurs only if the electrons undergo elastic collisions.
- Thus, when $V_A = 4.9 \times n$ Volts ($n=1, 2, \dots$) the electrons undergo inelastic collisions with Hg atoms.
- In inelastic collision kinetic energy of electrons is transferred to the internal energy of Hg atoms. Hg atoms absorb the energy of electrons.



Quantum Mechanics

Wave-Particle Duality

Light: wave nature (diffraction, refraction, interference....)
particle nature (photoelectric effect, Compton effect)
└─ called “photon”

Electron: particle nature ($m_0, q \dots$)
wave nature (diffraction, refraction, interference...)

de Broglie's matter wave: All particles have the properties of wave.

$$\lambda = \frac{h}{mv} \quad \longrightarrow \quad p = \frac{h}{\lambda} \quad \text{de Broglie's hypothesis}$$

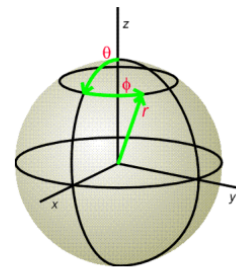
Wave Functions

Wavefunctions

- A particle **trajectory** is a classical concept.
- In Quantum Mechanics, a “particle” (e.g. an electron) does not follow a definite trajectory $\{\mathbf{r}(t), \mathbf{p}(t)\}$, but rather it is best described as being **distributed through space like a wave**.

Definitions

- **Wavefunction** (ψ) – a wave representing the spatial distribution of a “particle”.
- e.g. electrons in an atom are described by a wavefunction centred on the nucleus.
- ψ is a function of the coordinates defining the position of the classical particle:
 - 1-D $\psi(x)$
 - 3-D $\psi(x,y,z) = \psi(\mathbf{r}) = \psi(r, \theta, \phi)$ (e.g. atoms)
- ψ may be time dependent – e.g. $\psi(x,y,z,t)$



The Importance of ψ

- ψ completely defines the system (e.g. electron in an atom or molecule).
- If ψ is known, we can determine any observable property (e.g. energy, vibrational frequencies, ...) of the system.
- QM provides the tools to determine ψ computationally, to interpret ψ and to use ψ to determine properties of the system.

To get full knowledge of a problem **POSITION** and **MOMENTUM** need to be known

For **QUANTUM particles** its Uncertain $\rightarrow \Delta x \Delta p_x \geq \frac{\hbar}{2}$

First Postulate Introduce a parameter called

$$\Psi(x)$$

- Δx and Δp , are not absolute values, they represents the accuracy of measurement.
- The state of a Quantum Object is completely specified by a Wave Function $\psi(x)$

$\Psi^*(x)\Psi(x)$ is the probability density.

For Quantum Particles probability gives full information.

Probability that the particle can be found in the interval dx

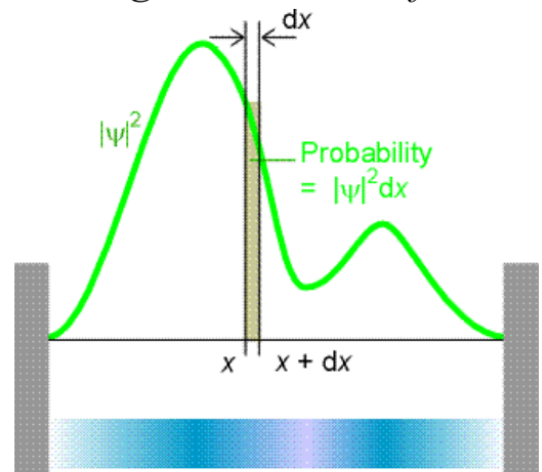
$$\Psi^*(x)\Psi(x)dx$$

A given wavefunction of the form $\Psi(x) = Af(x)$ needs to get *normalized such that total Probability = 1*, using the condition for **NORMALIZATION**

$$\int_{all} \Psi^*(x)\Psi(x)dx = 1$$

Suitable A is obtained

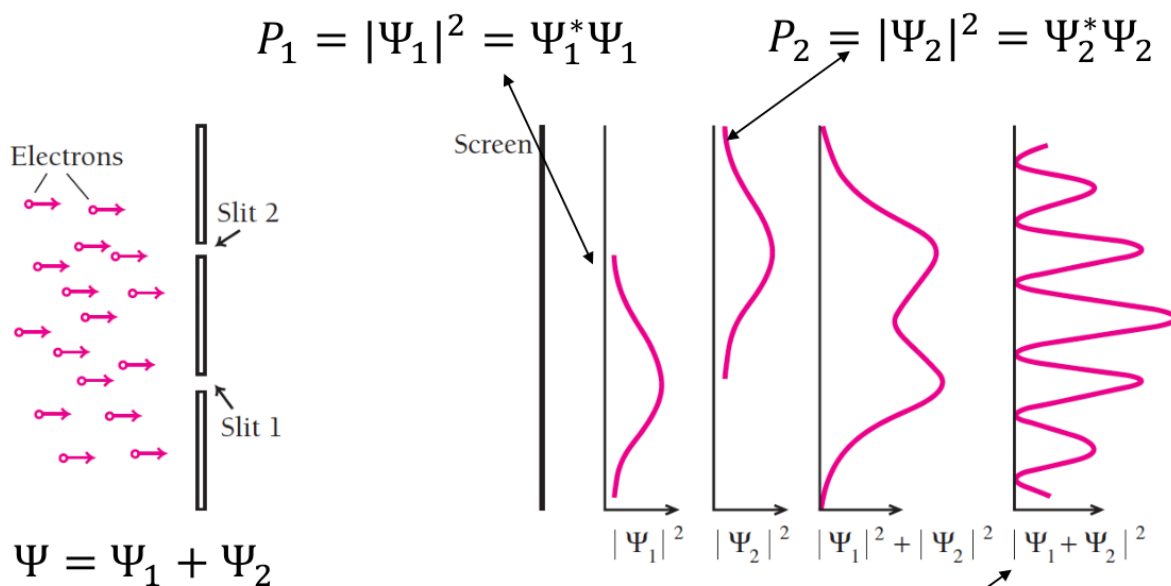
Probability must be real and positive
($0 \leq P \leq 1$).



Linearity and Superposition

Let $\Psi_1(x)$ and $\Psi_2(x)$ represent the two wave functions of a quantum system, then the linear superposition will also be a valid wavefunction for the same quantum system.

$$\Psi(x) = a_1\Psi_1(x) + a_2\Psi_2(x)$$



$$\begin{aligned} P &= |\Psi|^2 = |\Psi_1 + \Psi_2|^2 = (\Psi_1^* + \Psi_2^*)(\Psi_1 + \Psi_2) \\ &= \Psi_1^*\Psi_1 + \Psi_2^*\Psi_2 + \Psi_1^*\Psi_2 + \Psi_2^*\Psi_1 \\ &= P_1 + P_2 + \Psi_1^*\Psi_2 + \Psi_2^*\Psi_1 \end{aligned}$$

The wave functions Ψ_1 and Ψ_2 add to produce the intensity at the screen, not the probability densities $|\Psi_1|^2$ and $|\Psi_2|^2$.

Expectation value

Let $\Psi(x)$ describe the wavefunction of a quantum system, average value of the position (x) is given by the expectation value integral:

$$\langle x \rangle = \int \Psi^*(x) x \Psi(x) dx$$

Expectation value of position of a quantum particle

Probability that the particle can be found in the interval dx

$$\Psi^*(x) \Psi(x) dx$$

What is the expectation value of the position of the particle ?

$$\langle x \rangle = \frac{P_1 x_1 + P_2 x_2}{P_1 + P_2}$$

$$\langle x \rangle = \frac{\int_{all} x \Psi^*(x) \Psi(x) dx}{\int_{all} \Psi^*(x) \Psi(x) dx}$$

One-electron Schrödinger equation

General Formulation

- (1) There exist a wavefunction, $\Psi(x, y, z, t)$, describing the dynamic behavior of the system. $\downarrow$ mathematically complex quantity
- (2) Ψ and $\nabla\Psi$ must be continuous, finite, and single-valued for all values of x, y, z , and t .
- (3) $\Psi^* \Psi dV = |\Psi|^2 dV$: the probability that the particle will be found in the spatial volume element dV .
- $$\int \Psi^* \Psi dV = 1: \text{ normalization}$$

Operators

Operators and Observables

- If ψ is the wavefunction representing a system, we can write:

$$\hat{Q}\psi = Q\psi$$

where Q – “observable” property of system (e.g. energy, momentum, dipole moment ...)

$\hat{Q}$ – operator corresponding to observable Q .

- This is an **eigenvalue equation** and can be rewritten as:

$$\hat{Q}(\psi) = Q \times \psi$$

operator Q acting on function ψ (eigenfunction) function ψ multiplied by a number Q (eigenvalue)

(Note: ψ can't be cancelled).

Examples: $d/dx (e^{ax}) = a e^{ax}$

Quantum Mechanical Position and Momentum Operators

- Operator for position in the x-direction is just multiplication by x

$$\hat{x}\psi = x\psi$$

- Operator for linear momentum in the x-direction: $\hat{p}_x = \left(\frac{\hbar}{i}\right) \left(\frac{d}{dx}\right)$

$$\hat{p}_x\psi = p_x\psi \Rightarrow \frac{\hbar}{i} \cdot \frac{d\psi}{dx} = p_x\psi$$

(solve first order differential equation $\Rightarrow \psi, p_x$).

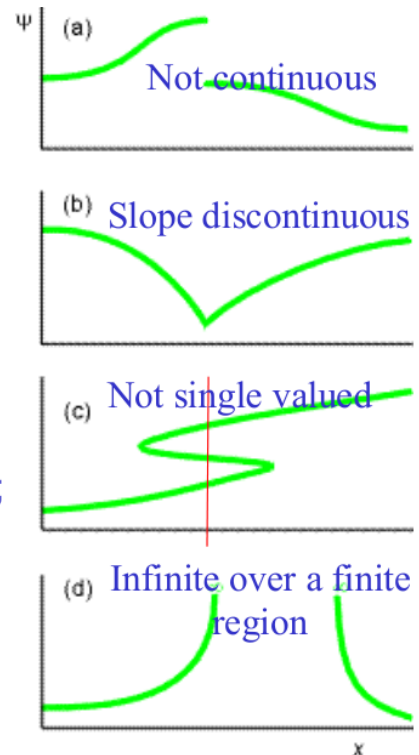
Physical properties		Operators	
Name of Operator	Observables	Operators	Symbols
Position	Position with x coordinate	x	$\hat{x}$
Momentum	x component of momentum	$-i\hbar \frac{\partial}{\partial x}$	$\widehat{p_x}$
Kinetic Energy	Kinetic Energy	$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$	$\hat{T}$
Potential Energy	Potential Energy	V(x)	$\hat{V}$
Total Energy (E)	Hamiltonian Operator (Time-independent)	$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$	$\hat{H}$
Total Energy (E)	Hamiltonian Operator (Time-dependent)	$-i\hbar \frac{\partial}{\partial t}$	$\hat{H}$

The Wave Function

The Wavefunction

The Born interpretation of ψ places restrictions on the form of the wavefunction:

- (a) ψ must be continuous (no breaks);
- (b) The gradient of ψ ($d\psi/dx$) must be continuous (no kinks);
- (c) ψ must have a single value at any point in space;
- (d) ψ must be finite everywhere;
- (e) ψ cannot be zero everywhere.



- Other restrictions (**boundary conditions**) depend on the exact system.
- These restrictions on ψ mean that only certain wavefunctions and $\therefore$ only certain energies of the system are allowed.

$\therefore$ Quantization of $\psi \Rightarrow$ Quantization of E

To find ψ and calculate the properties (observables) of a system:

1. Construct relevant operator $\hat{Q}$
2. Set up equation $\hat{Q}\psi = Q\psi$
3. Solve equation $\Rightarrow$ allowed values of ψ and Q .

Schrodinger Equation

The Schrödinger Equation (1926)

- The central equation in Quantum Mechanics.
- Observable = total energy of system.

Schrödinger Equation $\hat{H}\psi = E\psi$ $\hat{H}$ Hamiltonian Operator
E Total Energy

where $\hat{H} = \hat{T} + \hat{V}$ and $E = T + V$.

- SE can be set up for any physical system.
- The form of $\hat{H}$ depends on the system.
- Solve SE $\Rightarrow \psi$ and corresponding E.

Examples

1. Particle Moving in 1-D $\psi(x)$

$$\hat{H}\psi = \hat{T}\psi + \hat{V}\psi = E\psi \longrightarrow -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right) + V(x)\psi = E\psi$$

- The form of $V(x)$ depends on the physical situation:
 - Finite well $V(x) = 0$ for all a range of x and finite for rest.
 - Infinite well $V(x) = 0$ for all a range of x and infinite for rest.
 - Harmonic oscillator $V(x) = \frac{1}{2}kx^2$

The Particle in a Box

Particle in 1-dimensional box

Time Independent Schrödinger Equation

$$\underbrace{\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2}}_{\text{KE}} + \underbrace{V(x)\psi}_{\text{PE}} = \underbrace{E\psi}_{\text{TE}}$$

• Infinite walls

Region I: $V(x)=\infty$
 Region II: $V(x)=0$
 Region III: $V(x)=\infty$

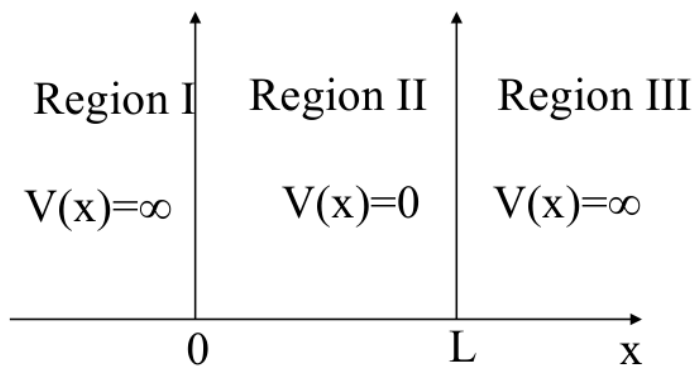
System
 Particle of mass m in 1-D box of length L .
 Position $x = 0 \rightarrow L$.

Particle cannot escape from box as PE $V(x)=\infty$ for $x = 0, L$ (walls).
 PE inside box: $V(x)=0$ for $0 < x < L$.

Restrictions on ψ

- In principle **Schrödinger Eqn.** has an infinite number of solutions.
- So far we have general solutions: any value of $\{\psi, E\}$.
- BUT** – due to the **Born interpretation of ψ** , only certain values of ψ are physically acceptable:
 - outside box ($x < 0, x > L$) $V = \infty \Rightarrow$ impossible for particle to be outside the box
 $\therefore |\psi|^2 = 0 \Rightarrow \psi = 0$ outside box.
 - ψ must be a continuous function
 $\Rightarrow$ **Boundary Conditions** $\psi = 0$ at $x = 0$
 $\psi = 0$ at $x = L$.

Applying boundary conditions: Restrictions on ψ



Region I and III: $\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \infty * \psi = E\psi \quad \Rightarrow |\psi|^2 = 0$

Region II:

$$\frac{d^2\Psi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \Psi(x)$$

$$\frac{d^2\Psi(x)}{dx^2} = -k^2 \Psi(x) \quad k^2 = \frac{2mE}{\hbar^2}$$

At $x = L$ $\psi = 0$

$$0 = A \sin kL$$

But $A \neq 0$

$$kL = n\pi$$

This is similar to the general differential equation:

$$\psi = A \sin kx + B \cos kx$$

Thus, wave function:

$$\psi_{II} = A \sin \frac{n\pi x}{L}$$

But what is 'A'?

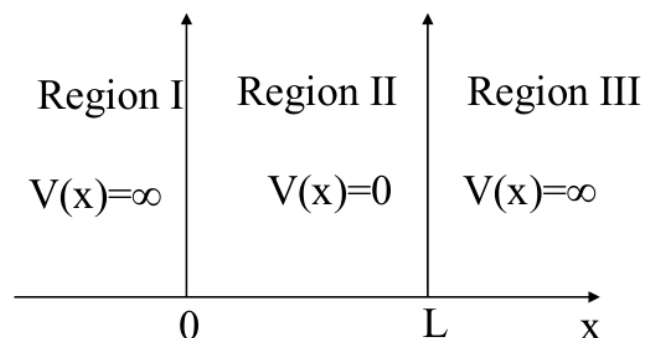
Applying boundary conditions:

At $x = 0 \rightarrow \psi = 0$

$$0 = A \sin 0k + B \cos 0k$$

$$0 = 0 + B * 1$$

$$\Rightarrow B = 0$$



Normalizing wave function:

$$\int_0^L (A \sin kx)^2 dx = 1$$

$$\sin^2 a = \frac{(1 - \cos(2a))}{2}$$

$$\int_0^L A^2 \left(\frac{1}{2} - \frac{\cos 2kx}{2} \right) dx = 1$$

$$|A|^2 \left[\frac{x}{2} - \frac{\sin 2kx}{4k} \right] \Big|_0^L = 1$$

$$|A|^2 \left[\frac{L}{2} - \frac{\sin 2 \frac{n\pi}{L} L}{4 \frac{n\pi}{L}} \right] = 1$$

$$k = \frac{n\pi}{L}$$

$$\sin 2\pi = 0$$

$$|A|^2 \left(\frac{L}{2} \right) = 1$$

$$|A| = \sqrt{\frac{2}{L}}$$

Thus normalized wave function is:

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Calculating Energy Levels:

$$k^2 = \frac{2mE}{\hbar^2}$$

$$E = \frac{k^2 \hbar^2}{2m}$$

$$E = \frac{k^2 h^2}{2m4\pi^2}$$

$$k = \frac{n\pi}{L}$$

$$(\hbar = \frac{h}{2\pi})$$

$$E = \frac{n^2 \pi^2}{L^2} \frac{h^2}{2m4\pi^2}$$

Thus Allowed Energies, E_n , are:

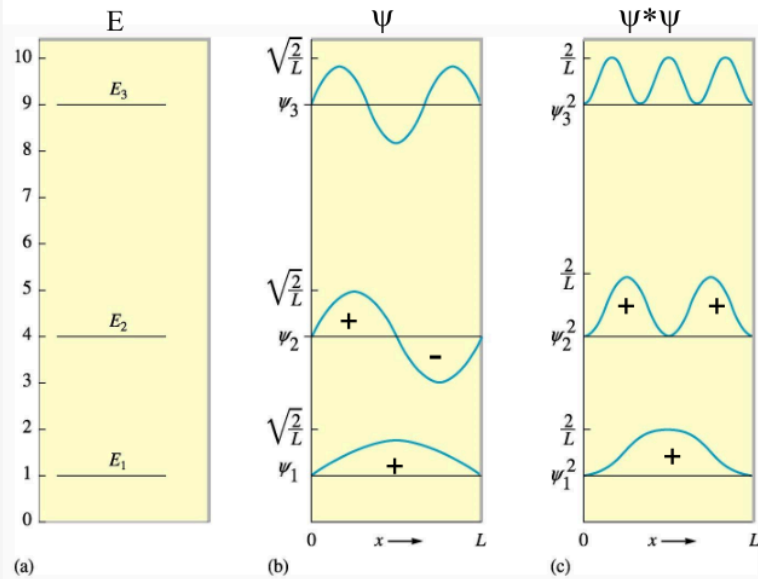
$$E_n = \frac{n^2 h^2}{8mL^2}$$

Particle in a 1-Dimensional Box

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

$$|\psi_{II}|^2 = \frac{2}{L} \left(\sin \frac{n\pi x}{L} \right)^2$$



Difference b/w adjacent energy levels:

$$\Delta E = E_{n+1} - E_n = (2n + 1)E_1$$

Expectation Value

Position

$$\psi_n = \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right)$$

$$\langle x \rangle = \int_0^L x \frac{2}{L} \sin^2 \left(\frac{n\pi x}{L} \right) dx$$

$$= \frac{2}{L} \int_0^L x \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi x}{L} \right) dx$$

$$= \frac{2}{L} \left[\frac{x^2}{4} - x \frac{\sin \left(\frac{2n\pi x}{L} \right)}{\left(\frac{4n\pi}{L} \right)} - \frac{\cos \left(\frac{2n\pi x}{L} \right)}{8 \left(\frac{n\pi}{L} \right)^2} \right]_0^L$$

$$= \frac{2}{L} \left[\frac{x^2}{4} \right]_0^L - \frac{2}{L} \frac{L^2}{8n^2\pi^2} \left[\frac{2n\pi x}{L} \sin \left(\frac{2n\pi x}{L} \right) + \cos \left(\frac{2n\pi x}{L} \right) \right]_0^L$$

Since $\sin(n\pi) = 0$ and $\cos(n\pi) = 1$, for all the values of n the expectation value of x is

$$\langle x \rangle = \frac{2}{L} \left(\frac{L^2}{4} \right) = \frac{L}{2}$$

Momentum

Momentum

$$\langle p_x \rangle = \int_0^L \left[\left[\frac{2}{L} \right]^{1/2} \sin\left(\frac{n\pi x}{L}\right) \right] \left(-\frac{i\hbar}{2\pi} \frac{d}{dx} \right) \left[\left[\frac{2}{L} \right]^{1/2} \sin\left(\frac{n\pi x}{L}\right) \right] dx$$

$$\langle p_x \rangle = -\frac{2\pi i \hbar}{L^2} \int_0^L \underbrace{\sin\left(\frac{n\pi x}{L}\right)}_{\text{odd}} \underbrace{\cos\left(\frac{n\pi x}{L}\right)}_{\text{even}} dx = 0$$

Also try with

$$\sin\left(\frac{n\pi x}{L}\right) = u$$

$$\frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) dx = du$$

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2\pi} \sin^2\left(\frac{n\pi x}{L}\right) = 0$$

Trigonometric Functions

Even Functions $f(-x) = f(x)$	Odd Functions $f(-x) = -f(x)$
$\cos(-x) = \cos x$ $\sec(-x) = \sec x$	$\sin(-x) = -\sin x$ $\csc(-x) = -\csc x$ $\tan(-x) = -\tan x$ $\cot(-x) = -\cot x$

Finite Potential Well

Finite Potential Well

Schrödinger Equation $\rightarrow$
$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + U(x)\psi(x) = E\psi(x)$$

I. $x < 0; U(x) = U$

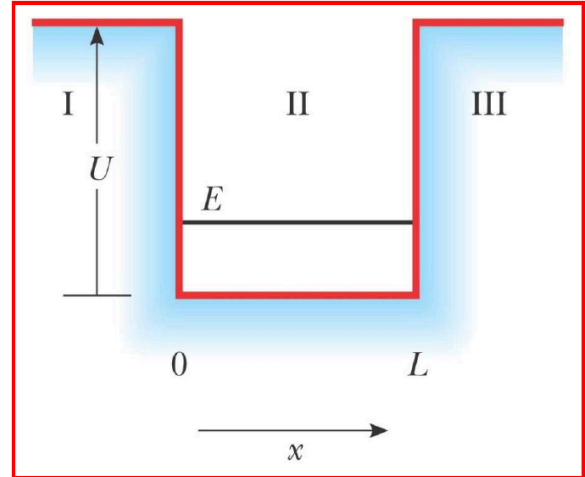
$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_I}{dx^2} + U\psi_I = E\psi_I$$

I. $0 < x < L; U(x) = 0$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_{II}}{dx^2} = E\psi_{II}$$

III. $x > L; U(x) = U$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_{III}}{dx^2} + U\psi_{III} = E\psi_{III}$$



We also assume that energy of the particle, E , is less than the “height” of the barrier, i.e.

$$E < U$$

Finite Potential Well: Regions I and III

- The Schrödinger equation for these regions is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U\psi = E\psi$$

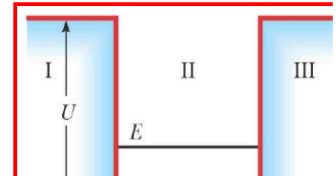
- It can be re-written as

$$\frac{d^2\psi}{dx^2} = \frac{2m(U - E)}{\hbar^2} \psi = C^2\psi, \quad \text{where } C^2 \equiv \frac{2m(U - E)}{\hbar^2}$$

- The general solution of this equation is

$$\psi(x) = Ae^{Cx} + Be^{-Cx}$$

– A and B are constants



- Requiring that wavefunction was finite at $x \rightarrow \infty$ and $x \rightarrow -\infty$, we can show that
- In region I, $B = 0$, and $\psi_I(x) = Ae^{Cx}$
 - This is necessary to avoid an infinite value for $\psi(x)$ for large negative values of x
- In region III, $A = 0$, and $\psi_{III}(x) = Be^{-Cx}$
 - This is necessary to avoid an infinite value for $\psi(x)$ for large positive values of x

The wavefunction and its derivative must be *single-valued* for all x

- At $x = 0$ and $x = L$ the wave function and its derivative have to be *continuous*

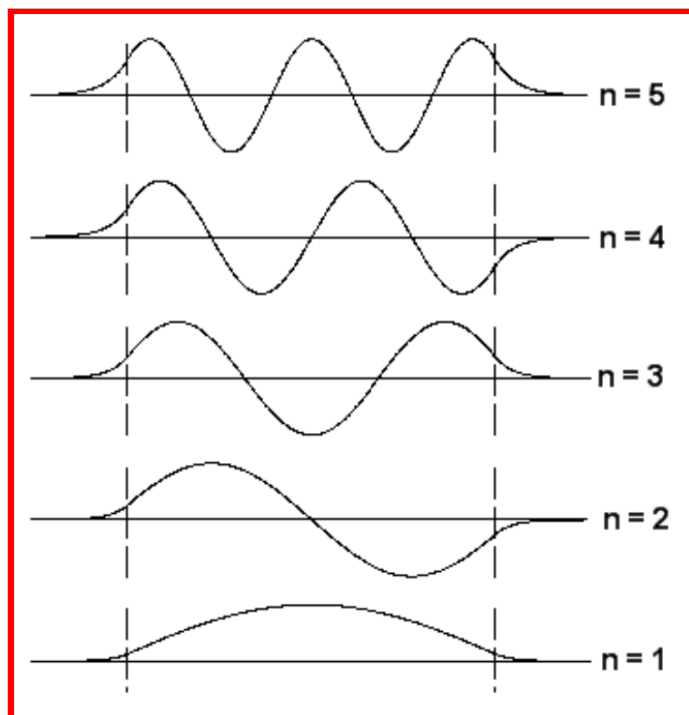
Thus, we have to equate “parts” of the wavefunction and its derivative at $x = 0, L$

- This, together with normalization condition, allows to determine the constants and the equation for energy of the particle

$$\begin{aligned}\psi_I(0) &= \psi_{II}(0) \\ \frac{d\psi_I}{dx}(0) &= \frac{d\psi_{II}}{dx}(0) \\ \psi_{II}(L) &= \psi_{III}(L) \\ \frac{d\psi_{II}}{dx}(L) &= \frac{d\psi_{III}}{dx}(L)\end{aligned}$$

$$\psi_{II}(x) = F \sin kx + G \cos kx$$

$$\psi_I(x) = Ae^{Cx}$$



$$\psi_{III}(x) = Be^{-Cx}$$

Graphical Results for $\psi(x)$

The particle penetrates the walls!

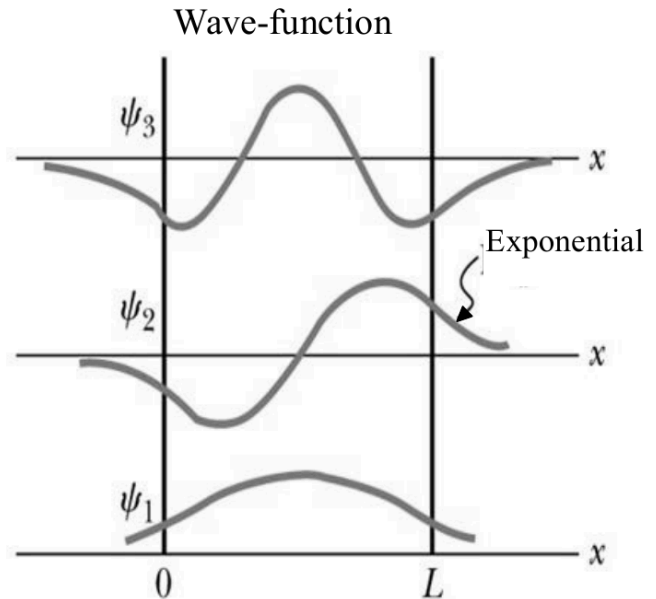
This violates classical physics!

The penetration depth is the distance outside the potential well where the probability decreases to about $1/e$. It's given by:

$$\delta x \approx \frac{1}{C} = \frac{\hbar}{\sqrt{2m(U - E)}}$$

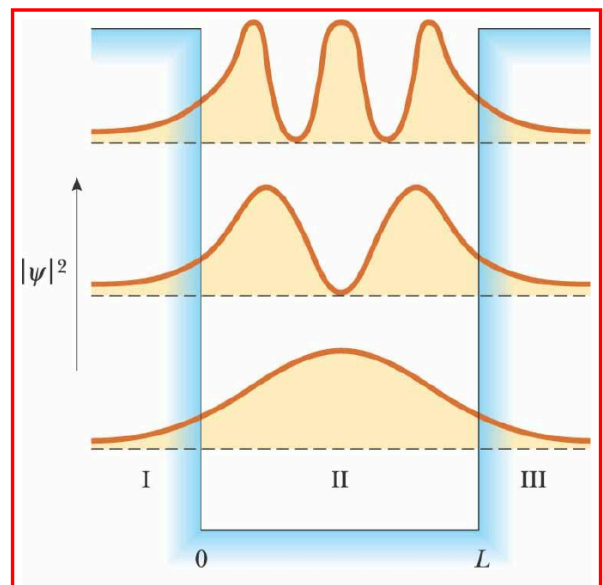
Note that the penetration distance is proportional to Planck's constant.
 $h = 10^{-34} \text{ Joule} - \text{seconds} \rightarrow$ usually an extremely tiny distance.

As $n \uparrow \rightarrow E \uparrow \rightarrow$ Penetration depth increases more and more.

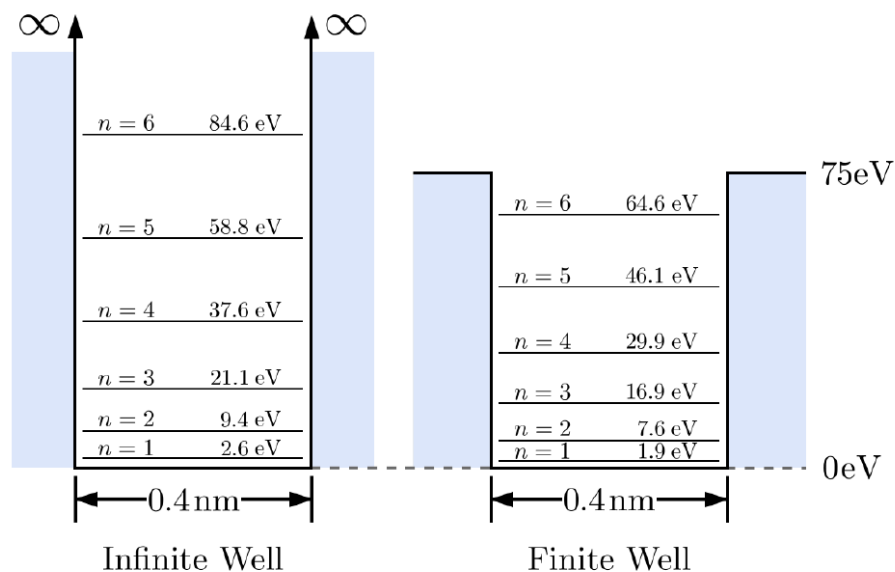


$$|\psi(x)|^2$$

- The probability densities for the lowest three states are shown
- The functions are smooth at the boundaries
- Outside the box, the probability to find the particle decreases exponentially, **but it is not zero!**



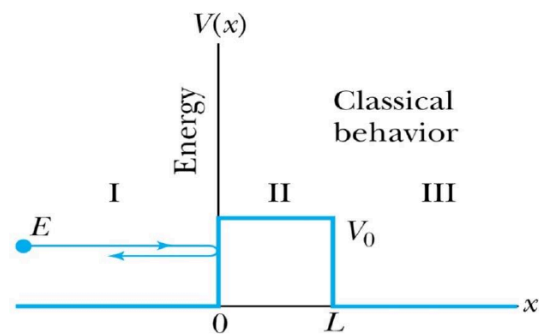
Comparison with infinite-well potential: The energy of state n is lower in the finite square well potential of the same width.



Tunneling

Consider a particle of energy E approaching a potential barrier of height V_0 , and the potential everywhere else is zero.

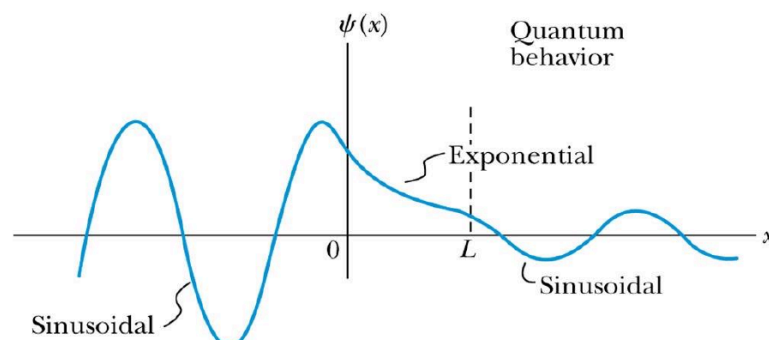
Also $E < V_0$.



The solutions in Regions I and III are sinusoids.

The wave function in region II becomes: $\psi_{II} = C e^{\alpha x} + D e^{-\alpha x}$

Enforcing continuity at the boundaries:



Tunneling is key in radioactivity.

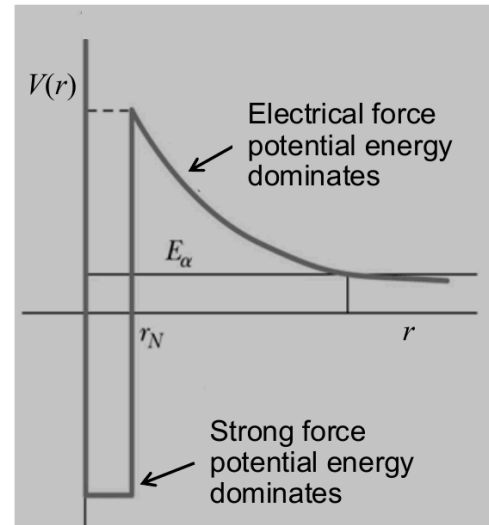
Radioactive Decay

Inside a nucleus, there's only positive charge, which is repulsive.

But inside the nucleus, the strong, short-range attractive nuclear force (~a square well) dominates the repulsive electrical force.

The potential barrier at the nuclear radius is several times greater than the energy of an alpha particle, E_α .

In quantum mechanics, however, the alpha particle can **tunnel** through the barrier.



α -particle potential energy curve

This is an excellent model for radioactive decay!

Harmonic Oscillator

Simple Harmonic Motion from the View of Quantum Mechanics

In order to know the wave function that describe the harmonic vibrational motion for the simple harmonic oscillator and the allowed energy levels, we have to solve the Schrödinger equation

$$\hat{H}\psi = E\psi$$

the Hamiltonian operator is given by

$$\hat{H} = \hat{T} + \hat{V}$$

the kinetic energy operator is given by

$$\hat{T} = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2}$$

notice that, we are assuming that the motion of the simple harmonic oscillator (e.g. the vibration of a diatomic molecule) is in one direction and we assumed that the motion is along the x - axis.

The potential energy operator

$$\hat{V} = \frac{1}{2} kx^2$$

Now substituting $\hat{T}$ and $\hat{V}$ into the one-dimensional Schrödinger equation gives $-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} \psi(x) = E \psi(x)$

After some effort, the eigenfunctions are $n = 0, 1, 2, \dots$

$$\psi_n(x) = N_n H_n\left(\frac{\alpha}{2} x\right) e^{-\frac{\alpha x^2}{2}}$$

where H_n is the Hermite polynomial of degree n .

α is defined as:

$$\alpha = \frac{2\pi\mu\nu}{\hbar}$$

N_n is the normalization constant which is defined as:

$$N_n = (2^n n!)^{-\frac{1}{2}} \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$$

Table 5.2 Some Hermite Polynomials

n	$H_n(y)$	α_n	E_n
0	1	1	$\frac{1}{2}\hbar\nu$
1	$2y$	3	$\frac{3}{2}\hbar\nu$
2	$4y^2 - 2$	5	$\frac{5}{2}\hbar\nu$
3	$8y^3 - 12y$	7	$\frac{7}{2}\hbar\nu$
4	$16y^4 - 48y^2 + 12$	9	$\frac{9}{2}\hbar\nu$
5	$32y^5 - 160y^3 + 120y$	11	$\frac{11}{2}\hbar\nu$

$$y = \sqrt{\alpha} x$$

The eigenvalues are

$$E_n = \left(n + \frac{1}{2}\right) \hbar\nu \quad n = 0, 1, 2, 3, \dots$$

where n = the vibrational quantum number.

IMPORTANT : The energy is restricted to the quantized values

Recall that,

$$\nu = \frac{\omega}{2\pi} \quad \text{and} \quad \omega = \sqrt{k/\mu} \Rightarrow \nu = \frac{1}{2\pi} \sqrt{k/\mu}$$

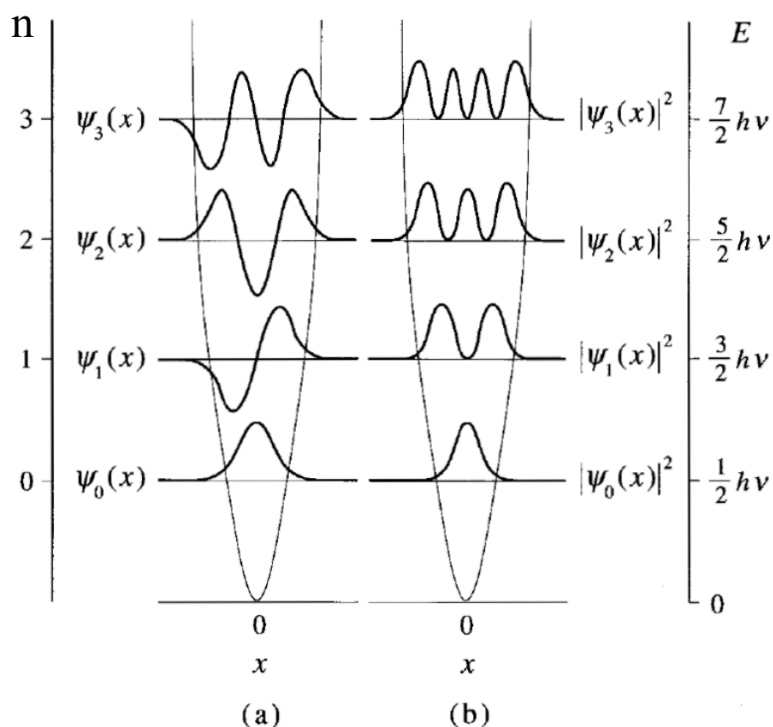
And

$$\hbar = \frac{h}{2\pi} \Rightarrow h = \hbar 2\pi$$

Therefore,

$$E_v = \hbar(k/\mu)^{\frac{1}{2}} \left(n + \frac{1}{2}\right)$$

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

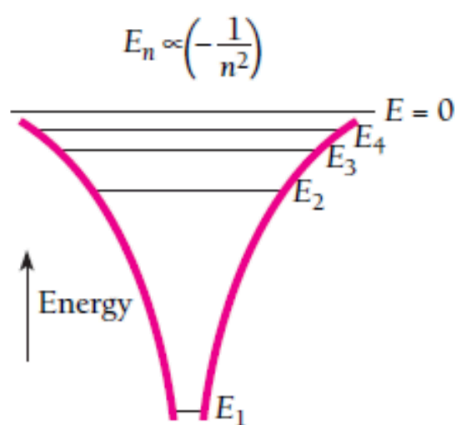


Levels are **equally spaced**:

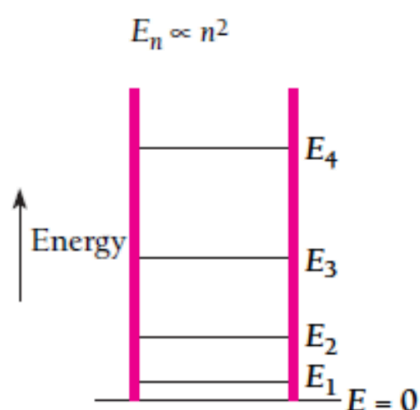
$$E_{n+1} - E_n = \hbar\omega = h\nu$$

(a) The normalized harmonic-oscillator wave functions.

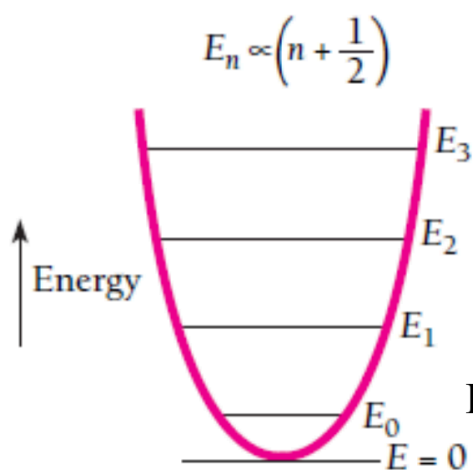
(b) The probability densities for a harmonic oscillator.



Coulombic Potential



Square well Potential



Harmonic Potential

Hydrogen Atom

SCHRÖDINGER'S EQUATION FOR THE HYDROGEN ATOM

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} (E - U) \psi = 0$$

Electric Potential energy $U = -\frac{e^2}{4\pi\epsilon_0 r}$.

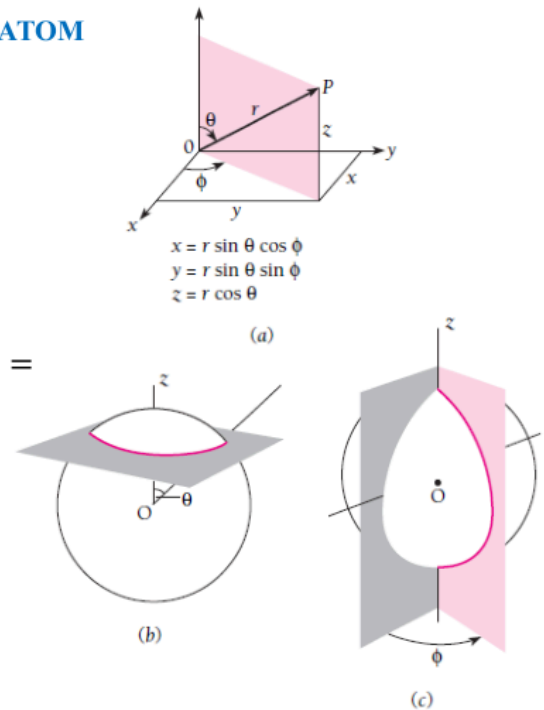
Negative sign $\rightarrow$ the potential is attractive

Spherical polar coordinates

r = length of the radius vector from origin O to point P = $\sqrt{x^2 + y^2 + z^2}$

θ = angle between the radius vector and the $+z$ axis
= zenith angle

$$= \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) = \cos^{-1} \left(\frac{z}{r} \right)$$



Figures:

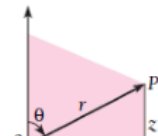
a) Spherical polar coordinates.

(b) polar angle $\rightarrow \theta \rightarrow \theta = 0$ means true north, and $\theta = \pi$ means true south.

(c) Azimuth angle $\rightarrow \phi \rightarrow$ the east or west angle from the x-axis.

ϕ = angle between the projection of the radius vector in the xy plane and the $+x$ axis, measured in the direction shown

$$= \text{azimuthal angle} = \tan^{-1} \frac{y}{x}$$



<https://planetmath.org/DerivationOfTheLaplacianFromRectangularToSphericalCoordinates>

In spherical polar coordinates Schrödinger's equation is written

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2m_e}{\hbar^2} (E - U) \psi = 0$$

Substitute the potential energy U and multiply the entire equation by $r^2 \sin^2 \theta$

Hydrogen Atom

$$\sin^2 \theta \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2m_e r^2 \sin^2 \theta}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) \psi = 0$$

Solving the polar Schrödinger equation gives three quantum numbers.
Q# $\rightarrow$ describe the electron in a hydrogen atom, in place of the single quantum number of the Bohr theory.

Hydrogen Atom wave function

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$$

$$\psi = R\Theta\Phi$$

solutions are **separable** into the product of three functions



it transforms a difficult 3D problem into three much easier 1D ones.

$$\begin{aligned}\frac{\partial \psi}{\partial r} &= \Theta\Phi \frac{\partial R}{\partial r} = \Theta\Phi \frac{dR}{dr} \\ \frac{\partial \psi}{\partial \theta} &= R\Phi \frac{\partial \Theta}{\partial \theta} = R\Phi \frac{d\Theta}{d\theta} \\ \frac{\partial^2 \psi}{\partial \phi^2} &= R\Theta \frac{\partial^2 \Phi}{\partial \phi^2} = R\Theta \frac{d^2 \Phi}{d\phi^2}\end{aligned}$$

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} + \frac{2m_e r^2 \sin^2 \theta}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = 0$$

The third term is a function of azimuth angle ϕ only, whereas the other terms are functions of r and θ only.

Rearrange

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{2m_e r^2 \sin^2 \theta}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = -\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2}$$

This eqn can be correct only if both sides of it are equal to the same constant, since they are functions of different variables .

It is convenient to call this constant m_l^2

The differential equation for the function Φ is therefore

$$-\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = m_l^2$$

Substitute m_l^2 for the RHS of topmost eqn on the slide, **divide the entire eqn by $\sin^2 \theta$** and rearrange the various terms

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2m_e r^2}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = \frac{m_l^2}{\sin^2 \theta} - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right)$$

The eqn has different variable appear on each side requiring both sides be equal to the same constant. This constant is called $l(l + 1)$. The equations for the functions Θ and R are :

$$\frac{m_l^2}{\sin^2\theta} - \frac{1}{\Theta \sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) = l(l + 1)$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2m_e r^2}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) = l(l + 1)$$

Equation for Φ $\frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} + m_l^2 \Phi = 0$

Equation for Θ $\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + \left[l(l + 1) - \frac{m_l^2}{\sin^2\theta} \right] \Theta = 0$

Equation for R $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{2m_e r^2}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) - \frac{l(l+1)}{r^2} \right] R = 0$

Each of these is an ordinary differential equation for a single function of a single variable.

Only the equation for R depends on the potential energy $U(r)$.

Additional NOTE

The $R(r)$ equation can be written in two equivalent ways

Either as an equation for $R(r)$, or for the reduced radial function $P(r)=rR(r)$

Define the reduced radial function: $P_{n\ell}(r) \equiv rR_{n\ell}(r)$

$$R_{n\ell}(r): \frac{1}{r^2} \cdot \frac{d}{dr} \left(r^2 \frac{dR_{n\ell}(r)}{dr} \right) - \frac{\ell(\ell + 1)R_{n\ell}(r)}{r^2} - \left(\frac{2m_e}{\hbar^2} \right) [U(r) - E_n] R_{n\ell}(r) = 0$$

OR

$$\text{For } P_{n\ell}(r): \frac{d^2 P_{n\ell}}{dr^2} - \left(\frac{2m_e}{\hbar^2} \right) \left[U(r) + \frac{\ell(\ell+1)\hbar^2}{2m_e r^2} - E_n \right] P_{n\ell}(r) = 0$$

$$R_{n\ell}(r) = \frac{P_{n\ell}(r)}{r}$$

Using $P(r)$ removes the first-derivative term and often makes the radial equation look like a 1D Schrödinger equation with an effective potential.

$R_{n\ell}(r)$: **Associated Laguerre functions.**

$$P_{n\ell}(r): \frac{d^2 P_{n\ell}}{dr^2} - \left(\frac{2m_e}{\hbar^2} \right) \left[U(r) + \frac{\ell(\ell+1)\hbar^2}{2m_e r^2} - E_n \right] P_{n\ell}(r) = 0$$

The first term,

$$U(r) = -\frac{e^2}{4\pi\epsilon_0 r},$$

is the ordinary attractive Coulomb potential. It pulls the electron toward the nucleus.

The second term,

$$\frac{\ell(\ell+1)\hbar^2}{2m_e r^2}$$

is called the **centrifugal potential** or **centrifugal barrier**.

It appears because the electron has angular momentum.

Classically, for a particle moving in a central potential,

$$E = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r).$$

The term

$$\frac{L^2}{2mr^2} \rightarrow \text{rotational kinetic energy.}$$

Quantum mechanically,

$$L^2 = l(l+1)\hbar^2,$$

so

$$\frac{L^2}{2m_e r^2} \rightarrow \frac{\hbar^2 l(l+1)}{2m_e r^2}.$$

Description of solution

- We will not go through the mathematics of the solution. We can only accept solutions for which the wave function is normalizable (does not blow up).
- Some radial solutions blow up at $r = 0$ or $r = \infty \rightarrow$ Hence discarded.
- The radial solutions $R(r) \rightarrow$ **Associated Laguerre functions**. $\rightarrow$ An exponential function $e^{-\alpha r}$, multiplied by a polynomial in r .
- The polar angular solutions $\Theta(\theta) \rightarrow$ **Legendre polynomial** $\rightarrow$ are polynomials containing powers of $\sin\theta$ and $\cos\theta$.
- The azimuthal solutions $\Phi(\phi) \rightarrow$ Periodic (have the same value at $\Phi(\phi)$ and $\Phi(\phi+2\pi)$, etc. They turn out to depend on $e^{im_l\phi}$, where m_l is an integer, either positive, negative or zero.

$$m_l = 0, \pm 1, \pm 2, \pm 3, \dots, \pm l$$

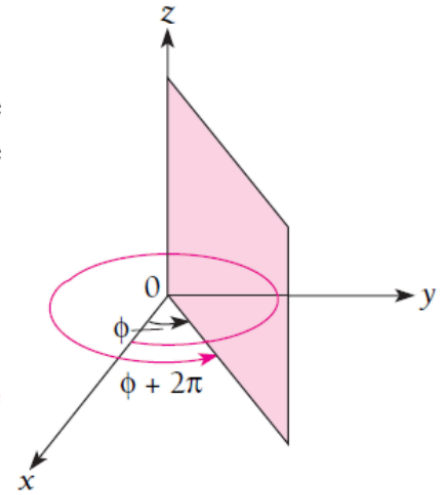


Figure 6.2 The angles ϕ and $\phi + 2\pi$ both identify the same meridian plane.

The solution for the radial part $R(r)$ of the hydrogen atom wave function Ψ also requires that a certain condition be fulfilled. This condition is that one of the negative values E_n (signifying that the electron is bound to the atom) specified by

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2} \frac{m_e e^4}{2n^2 \hbar^2} = \frac{E_1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}$$

$$n = 1, 2, 3, \dots$$

The same formula for the energy levels of the hydrogen atom that Bohr obtained.

Another condition that must be obeyed in order to solve the radial eqn is that n k/a the **Principal Q#**

$$n \geq (l + 1)$$

This requirement may be expressed as a condition on l in the form

$$l = 0, 1, 2, 3, \dots, (n - 1)$$

Principal quantum number

$$n = 1, 2, 3, \dots$$

Orbital quantum number

$$l = 0, 1, 2, \dots, (n - 1)$$

Magnetic quantum number

$$m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

n	l	m_l	$\Phi(\phi)$	$\Theta(\theta)$	$R(r)$	$\psi(r, \theta, \phi)$
1	0	0	$\frac{1}{\sqrt{2\pi}}$	$\frac{1}{\sqrt{2}}$	$\frac{2}{a_0^{3/2}} e^{-r/a_0}$	$\frac{1}{\sqrt{\pi} a_0^{3/2}} e^{-r/a_0}$
2	0	0	$\frac{1}{\sqrt{2\pi}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2\sqrt{2} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$	$\frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$
2	1	0	$\frac{1}{\sqrt{2\pi}}$	$\frac{\sqrt{6}}{2} \cos \theta$	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\frac{1}{4\sqrt{2\pi} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \cos \theta$
2	1	± 1	$\frac{1}{\sqrt{2\pi}} e^{\pm i\phi}$	$\frac{\sqrt{3}}{2} \sin \theta$	$\frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\frac{1}{8\sqrt{\pi} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \sin \theta e^{\pm i\phi}$

$$a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 5.292 \times 10^{-11} \text{ m}$$

PRINCIPAL QUANTUM NUMBER

Quantization of energy → n is quantized

ORBITAL QUANTUM NUMBER

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{2m_e}{\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0 r} + E \right) - \frac{l(l+1)}{r^2} \right] R = 0 \quad U = -\frac{e^2}{4\pi\epsilon_0 r}$$

$$E = KE_{radial} + KE_{orbital} + U = KE_{radial} + KE_{orbital} - \frac{e^2}{4\pi\epsilon_0 r}$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2m_e}{\hbar^2} \left[KE_{radial} + KE_{orbital} - \frac{\hbar^2 l(l+1)}{2m_e r^2} \right] R = 0$$

If the last two terms in the square brackets of the equation cancel each other out, we have a differential equation for $R(r)$ that involves functions of the radius vector r exclusively.

$$KE_{orbital} = \frac{\hbar^2 l(l+1)}{2m_e r^2}$$

Since the orbital kinetic energy of the electron and the magnitude of its angular momentum are

$$KE_{orbital} = \frac{1}{2} m_e v_{orbital}^2 \qquad L = m_e v_{orbital} r$$

The orbital kinetic energy

$$KE_{orbital} = \frac{L^2}{2m_e r^2}$$

Compare above with $KE_{orbital} = \frac{\hbar^2 l(l+1)}{2m_e r^2}$

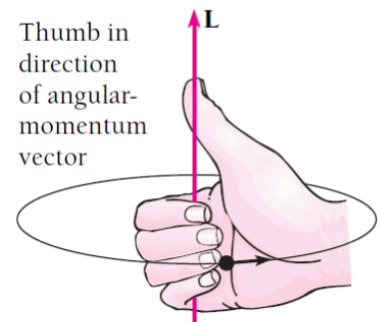
$$\frac{L^2}{2m_e r^2} = \frac{\hbar^2 l(l+1)}{2m_e r^2}$$

Electron angular momentum $L = \hbar \sqrt{l(l+1)}$
 $l = 0, 1, 2, 3, \dots \dots (n-1)$

Like total energy E , *angular momentum is both conserved and quantized.*

MAGNETIC QUANTUM NUMBER

- The orbital quantum number l determines the magnitude L of the electron's angular momentum.
- Angular momentum is a vector quantity, meaning both its magnitude and direction must be specified.
- The vector L is perpendicular to the plane of rotational motion, following the right-hand rule.
- For a hydrogen atom, the direction in space becomes significant because an electron revolving around the nucleus acts as a minute current loop, producing a magnetic field like a magnetic dipole.
- An electron with angular momentum interacts with an external magnetic field B .
- The magnetic quantum number m_l specifies the direction of L by determining its component along the direction of the magnetic field.
- This phenomenon is called space quantization.



Fingers of right hand in direction of rotational motion

Figure 6.3 The right-hand rule for angular momentum.

If the magnetic-field direction be parallel to the z axis, the component of $\mathbf{L}$ in z direction is

Space quantization $L_z = m_l \hbar \quad m_l = 0, \pm 1, \pm 2, \dots, \pm l$

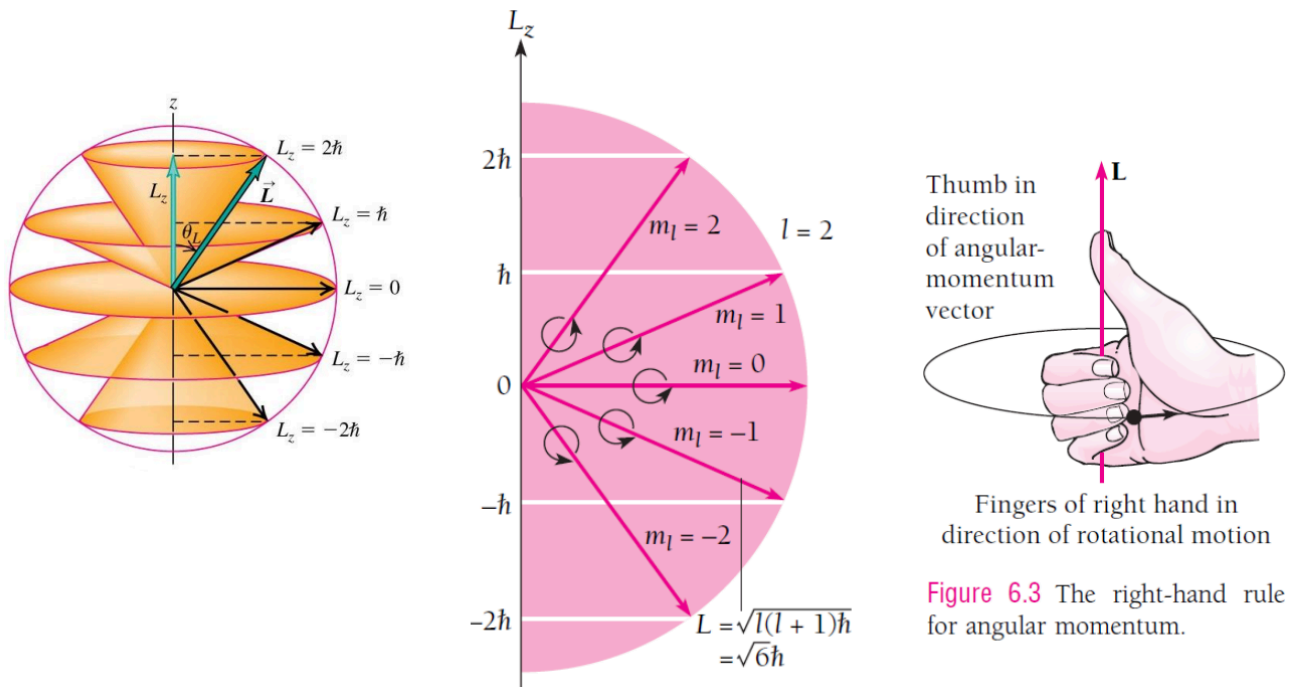


Figure 6.3 The right-hand rule for angular momentum.

ELECTRON PROBABILITY DENSITY

The probability density $|\Psi|^2$ that corresponds to the electron wave function $\Psi = R\Theta\Phi$ in the hydrogen atom is

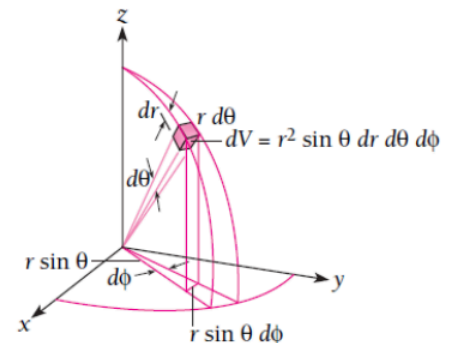
$$|\Psi|^2 = |R|^2 |\Theta|^2 |\Phi|^2$$

Probability of Finding the Electron

$$dP = \Psi^*(r, \theta, \phi) \Psi(r, \theta, \phi) dV$$

The differential volume element in spherical polar coordinates is

$$dV = r^2 \sin \theta dr d\theta d\phi$$



$$P(r)dr = r^2 R^*(r) R(r) dr \int_0^\pi |\Theta|^2 \sin \theta d\theta \int_0^{2\pi} |\Phi|^2 d\phi = r^2 |R|^2 dr$$

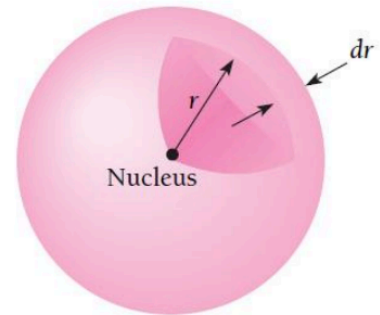
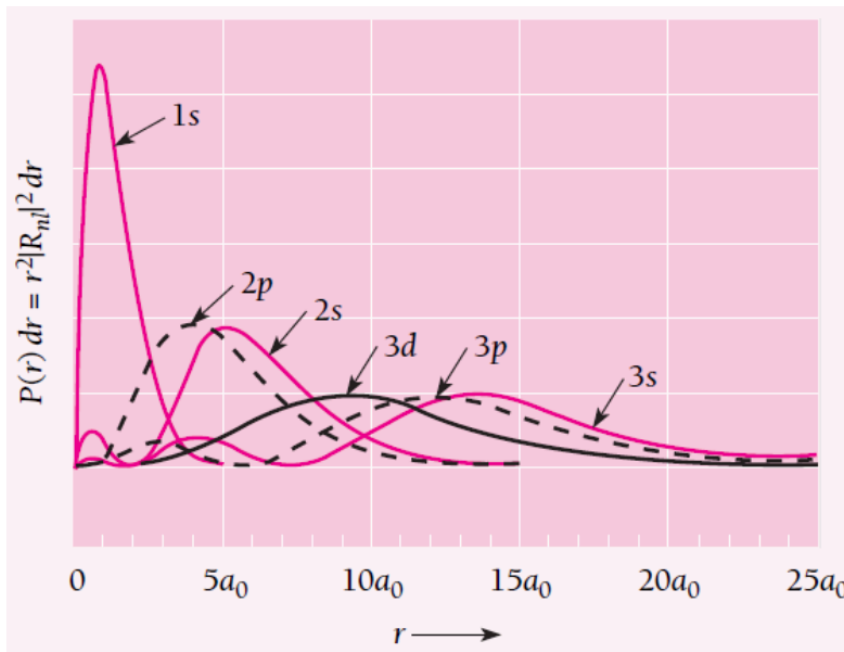
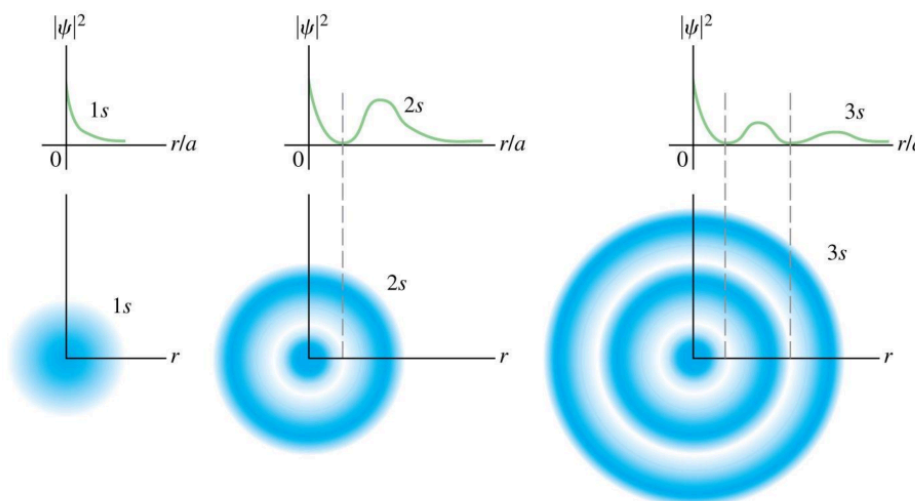


Figure 6.10 The probability of finding the electron in a hydrogen atom in the spherical shell between r and $r + dr$ from the nucleus is $P(r) dr$.

- The **most probable** value of r for a **1s electron** is a_0 , the orbital radius of a ground-state electron in the Bohr model.
- The **average** value of r for a **1s electron** is $1.5a_0$.

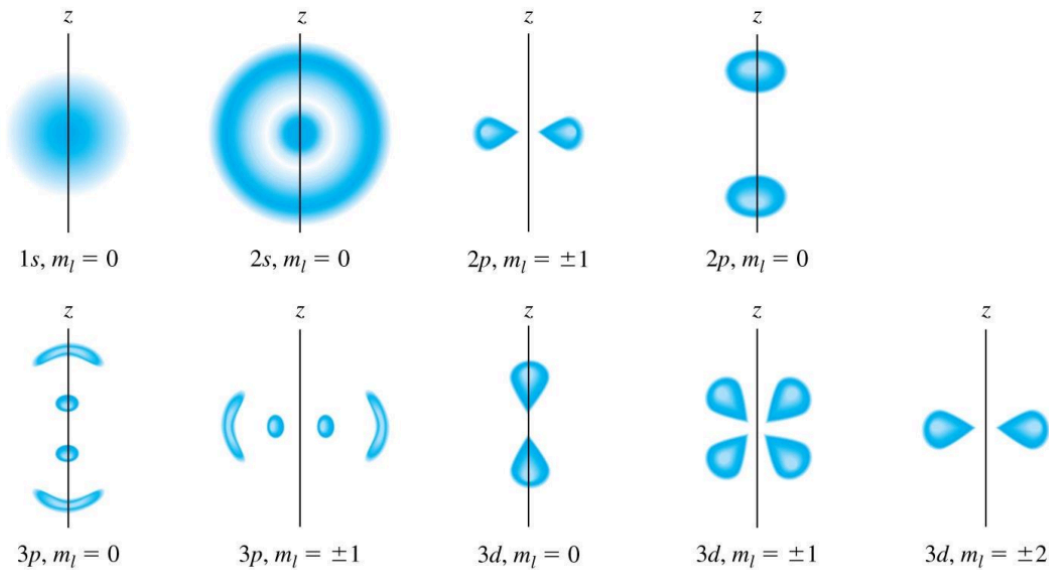
The hydrogen atom: Probability distributions I

- States of the hydrogen atom with $l = 0$ (zero orbital angular momentum) have spherically symmetric wave functions that depend on r but not on θ or ϕ . These are called s states.
- Figure below shows the electron probability distributions for three of these states.



The hydrogen atom: Probability distributions II

- States of the hydrogen atom with nonzero orbital angular momentum, such as p states ($l = 1$) and d states ($l = 2$), have wave functions that are *not* spherically symmetric.
- Figure below shows the electron probability distributions for several of these states, as well as for two spherically symmetric s states.



Number of nodes

- The total number of nodes in a hydrogen orbital is $n - 1$.
- The number of angular nodes is ℓ .
- The number of radial nodes is $n - \ell - 1$.
- Therefore, total nodes = $(n - \ell - 1) + \ell = n - 1$.

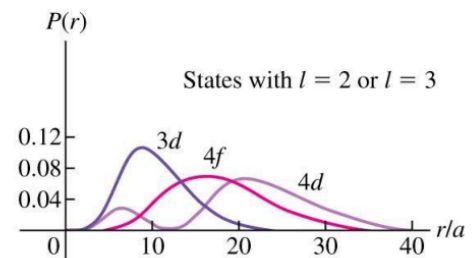
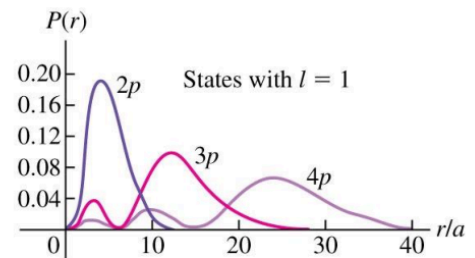
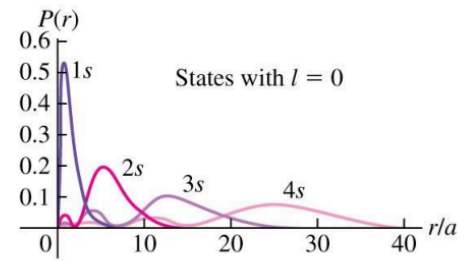
The hydrogen atom: Degeneracy

- States with different quantum numbers l and n are often referred to with letters as follows:

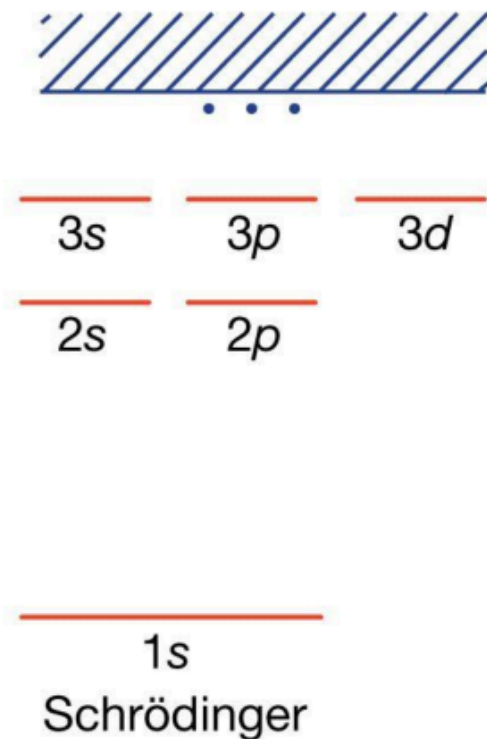
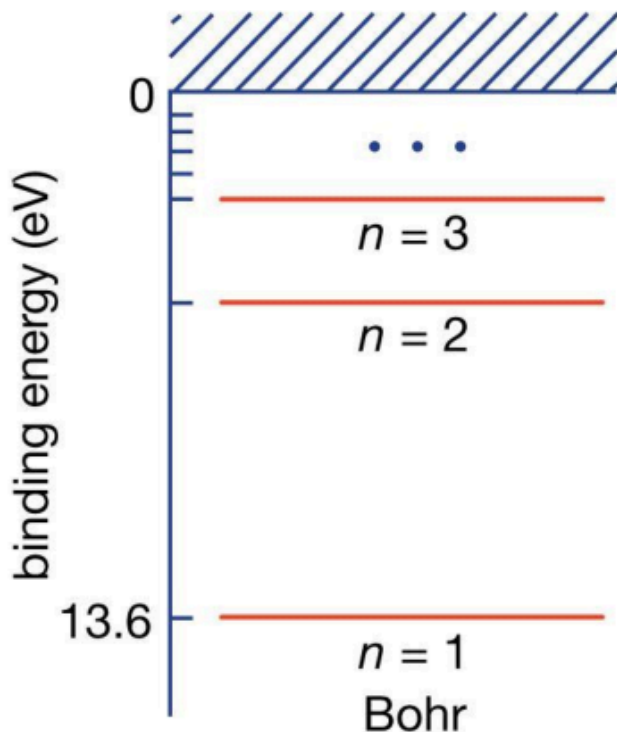
l value	letter
0	s
1	p
2	d
3	f
4	g
5	h

n value	shell
1	K
2	L
3	M
4	N

- Hydrogen atom states with **the same value of n but different values of l and m_l are degenerate** (have the same energy).
- Figures $\rightarrow$ radial probability distribution for states with $l = 0, 1, 2, 3$ and different values of $n = 1, 2, 3, 4$.



Visualising orbitals $\rightarrow$ <https://www.falstad.com/qmatom/>



RADIATIVE TRANSITIONS

Bohr postulated that the frequency of the radiation emitted by an atom dropping from an energy level E_m to a lower level E_n is

$$\nu = \frac{E_m - E_n}{h}$$

the time-dependent wave function Ψ_n of an electron in a state of quantum number n and energy E_n is the product of a time-independent wave function ψ_n and a time-varying function whose frequency is

$$\nu_n = \frac{E_n}{h}$$

$$\Psi_n = \psi_n e^{-\left(\frac{iE_n}{\hbar}\right)t} \qquad \Psi_n^* = \psi_n^* e^{+\left(\frac{iE_n}{\hbar}\right)t}$$

The expectation value $\langle x \rangle$ of the position of such an electron is

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{\infty} x \Psi_n^* \Psi_n dx = \int_{-\infty}^{\infty} x \psi_n^* \psi_n e^{[(iE_n/\hbar) - (iE_n/\hbar)]t} dx \\ &= \int_{-\infty}^{\infty} x \psi_n^* \psi_n dx \end{aligned}$$

The expectation value $\langle x \rangle \rightarrow$ constant in time since Ψ_n and Ψ_n^* are, by definition, functions of position only.

The electron does not oscillate, and no radiation occurs. Thus quantum mechanics predicts that a system in a specific quantum state does not radiate, as observed.

An electron that shifts from one energy state to another

The wave function Ψ_n of an electron that can exist in both states n and m

$$\Psi = a\Psi_n + b\Psi_m$$

a^*a = probability that the electron in state n

b^*b = probability that the electron in state m

$$a^*a + b^*b = 1$$

Initially $a=1$ and $b=0$;

when the electron is in the excited state $\rightarrow a=0$ and $b=1$; and ultimately $a=1$ and $b=0$ once more.

While the electron is in either state, there is no radiation, but when it is in the midst of the transition from m to n (that is, when both a and b have nonvanishing values), electromagnetic waves are produced.

The expectation value $\langle x \rangle$ that corresponds to the composite wave function

$$\begin{aligned}\langle x \rangle &= \int_{-\infty}^{\infty} x(a^*\Psi_n^* + b^*\Psi_m^*)(a\Psi_n + b\Psi_m) dx \\ &= \int_{-\infty}^{\infty} x(a^2\Psi_n^*\Psi_n + b^*a\Psi_m^*\Psi_n + a^*b\Psi_n^*\Psi_m + b^2\Psi_m^*\Psi_m) dx\end{aligned}$$

let $a^*a=a^2$ and $b^*b=b^2$.

The first and last integrals do not vary with time, so the second and third integrals are the only ones able to contribute to a time variation in $\langle x \rangle$.

$$\begin{aligned}\langle x \rangle = & a^2 \int_{-\infty}^{\infty} x \psi_n^* \psi_n dx + b^* a \int_{-\infty}^{\infty} x \psi_m^* e^{+(iE_m/\hbar)t} \psi_n e^{-(iE_n/\hbar)t} dx \\ & + a^* b \int_{-\infty}^{\infty} x \psi_n^* e^{+(iE_n/\hbar)t} \psi_m e^{-(iE_m/\hbar)t} dx + b^2 \int_{-\infty}^{\infty} x \psi_m^* \psi_m dx\end{aligned}$$

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \text{and} \quad e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\begin{aligned}\cos\left(\frac{E_m - E_n}{\hbar}\right)t \int_{-\infty}^{\infty} x [b^* a \psi_m^* \psi_n + a^* b \psi_n^* \psi_m] dx \\ + i \sin\left(\frac{E_m - E_n}{\hbar}\right)t \int_{-\infty}^{\infty} x [b^* a \psi_m^* \psi_n - a^* b \psi_n^* \psi_m] dx\end{aligned}$$

The real part of this result varies with time as

$$\cos\left(\frac{E_m - E_n}{\hbar}\right)t = \cos 2\pi\left(\frac{E_m - E_n}{h}\right)t = \cos 2\pi\nu t$$

The electron's position therefore oscillates sinusoidally at the frequency

$$\nu = \frac{E_m - E_n}{h}$$

- When the electron is in state n or state m the expectation value of the electron's position is constant.
- When the electron is undergoing a transition between these states, its position oscillates with the frequency .
- Such an electron, of course, is like an electric dipole and radiates electromagnetic waves of the same frequency .
- This result is the same as that postulated by Bohr and verified by experiment.

Selection Rules

The general condition necessary for an atom in an excited state to radiate is that the integral

$$\int_{-\infty}^{\infty} x \psi_n \psi_m^* dx \quad \text{not be zero}$$

Integral is finite $\rightarrow$ **allowed transitions**

Integral is zero $\rightarrow$ **forbidden transitions**

In the case of the hydrogen atom, three quantum numbers are needed to specify the initial and final states involved in a radiative transition. If the principal, orbital, and magnetic quantum numbers of the initial state are n', l', m'_l , respectively, and those of the final state are n, l, m_l , and u represents either the x, y , or z coordinate, the condition for an allowed transition is

Allowed transitions

$$\int_{-\infty}^{\infty} u \psi_{n,l,m_l} \psi_{n',l',m'_l}^* dV \neq 0$$

Since the wave functions Ψ_{n,l,m_l} for the hydrogen atom are known, the allowed transitions can be evaluated for $u = x$, $u = y$, and $u = z$ for all pairs of states differing in one or more quantum numbers.

When this is done, it is found that the only transitions between states of different n that can occur are those in which the orbital quantum number l changes by 1 or -1 and the magnetic quantum number m_l does not change or changes by 1 or -1 .

Selection Rules

$\Delta n =$ not restricted

$\Delta l = \pm 1$ (For all polarization of light)

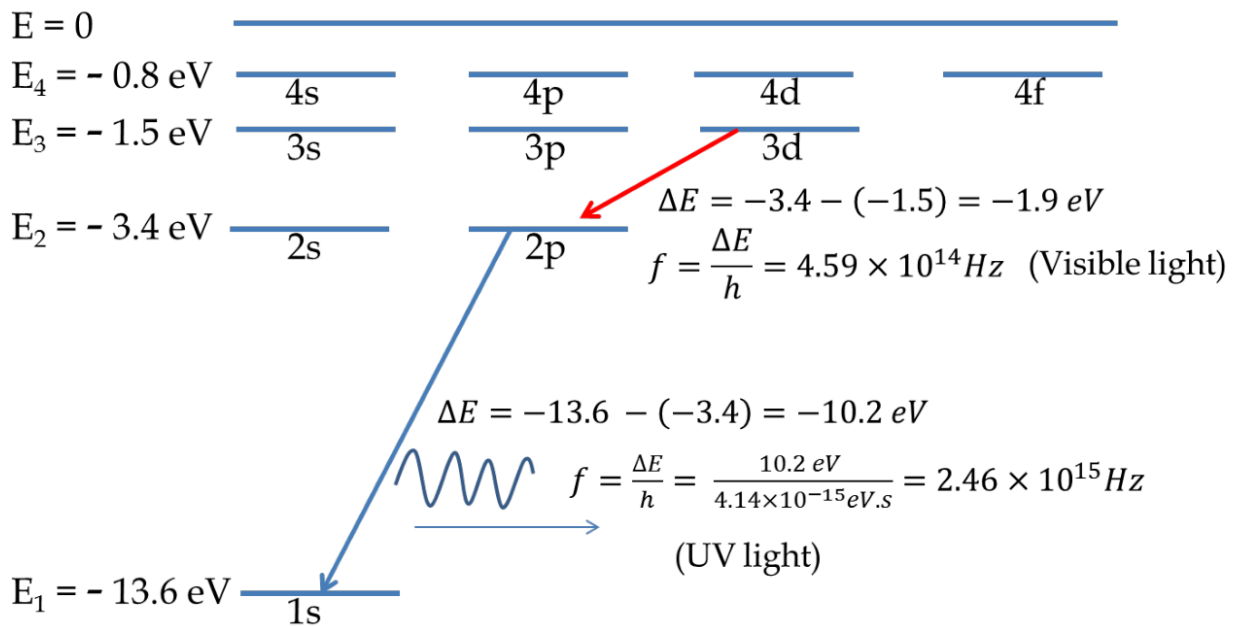
$\Delta m_l = 0$ (For linearly polarization of light)

$\Delta m_l = \pm 1$ (For circular polarization of light)

Excitation energy, eV



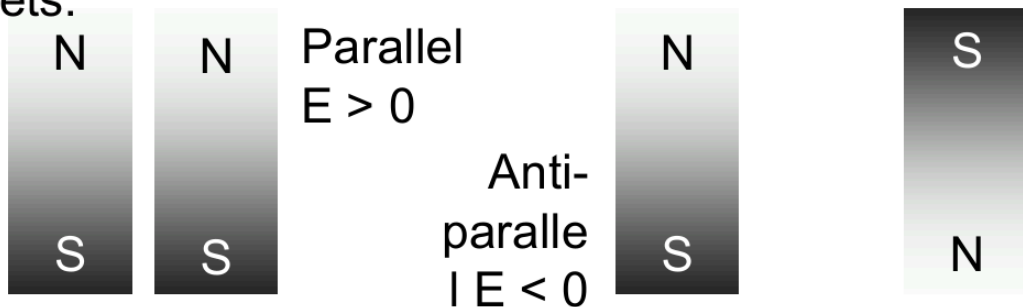
2s is a metastable state for hydrogen atom



Zeeman Effect

The Hydrogen Atom in a Magnetic Field

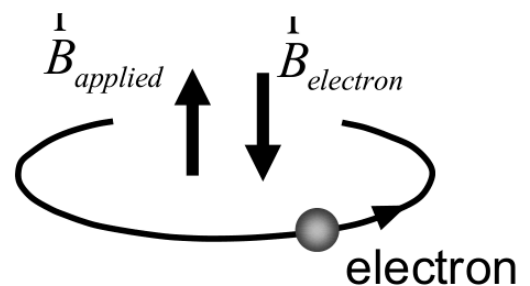
Consider pairs of magnets.



So, parallel magnets have more energy than anti-parallel magnets.

An electron orbiting around a nucleus generates a magnetic field.

So, an H atom and an applied magnetic field are two magnets.



In 1896, the Dutch physicist Pieter Zeeman showed that spectral lines emitted by atoms in a **magnetic field** split into multiple energy levels.

Think of an electron as an orbiting circular current loop of $I = dq/dt$ around the nucleus.

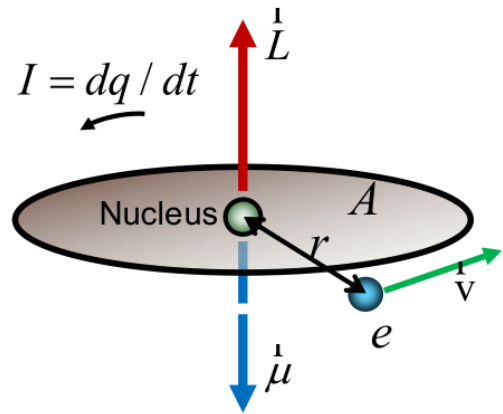
If the period is $T = 2\pi r/v$, then:

$$I = -e/T = -e/(2\pi r/v) = -ev/(2\pi r)$$

The current loop has a magnetic moment $\mu = IA$:

$$= [-ev/(2\pi r)]\pi r^2 = (-e/2m) mrv \quad \vec{\mu} = -\frac{e}{2m} \vec{L}$$

where $L = mvr$ is the magnitude of the orbital angular momentum.



ZEEMAN EFFECT

H atom in an external magnetic field $\rightarrow$ semiclassical model

the electron motion is described by a classical circular orbit, while the angular momentum is given by the quantum mechanical expression

$$|l| = \sqrt{l(l+1)} \hbar$$

An electron $\rightarrow -e$, moving with the velocity $\rightarrow v$,
the circular frequency $\rightarrow v = v/2\pi r$ on a circle $r \rightarrow$ radius
The generated electric current

$$I = -ev = -\frac{ev}{2\pi r} \quad \text{velocity}$$

The current causes a magnetic moment

$$\mu = IA = I\pi r^2 \hat{n}$$

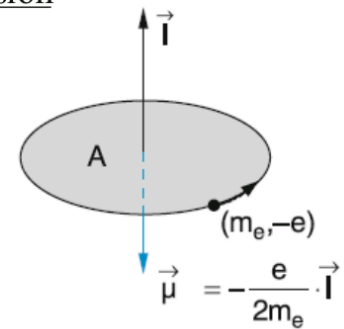
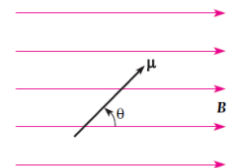
Where $A = \pi r^2 \hat{n}$ the area vector perpendicular to the plane of the motion

The angular momentum of the circulating electron is

$$l = r \times p = m_e r v \hat{n}$$

$$\mu = -\frac{e}{2m_e} l$$

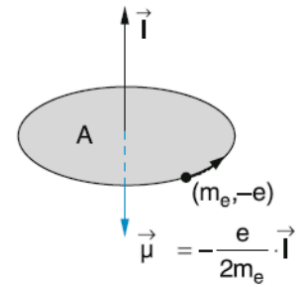
Since μ is proportional to l , the orbital magnetic moment is often labeled μ_l .



In an external magnetic field the potential energy of a magnetic dipole with magnetic moment μ is

$$E_{pot} = -\mu \cdot \mathbf{B} \quad \mu = -\frac{e}{2m_e} \mathbf{l}$$

$$= +\frac{e}{2m_e} \mathbf{l} \cdot \mathbf{B}$$

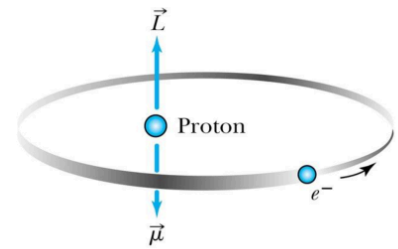


When the magnetic field points into the z -direction ($\mathbf{B} = \{0, 0, B_z = B\}$), because of $l_z = m\hbar$,

$$E_{pot} = +\frac{e\hbar}{2m_e} mB$$

where m (the projection of $\mathbf{l}$ onto the z -axis) is called the magnetic quantum number, It can take values $-l \leq m \leq +l$.

$$\mu_B = \frac{e\hbar}{2m_e} = 9.274015 \times 10^{-24} \text{ J/T} \rightarrow \text{Bohr magneton}$$



The additional energy caused by the magnetic field as

$$\Delta E_m = \mu_B mB$$

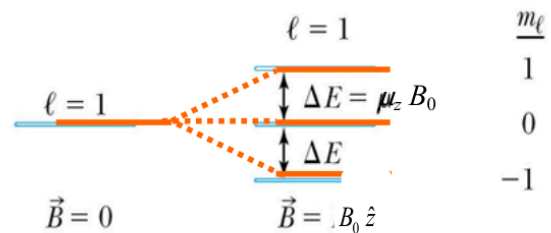
This gives for the energies of the hydrogen atomic states in an external magnetic field

$$E_{n,l,m} = E_{Coul}(n) + \mu_B mB$$

The $2l + 1$ *m-sublevels* that are degenerate without magnetic field split into $2l + 1$ *equidistant* Zeeman components with energies between

$$E_n^{min} = E_{coul}(n) - \mu_B mB$$

$$E_n^{max} = E_{coul}(n) + \mu_B mB$$



Zeeman splitting of a level with $l=1$ in a homogeneous magnetic field (normal Zeeman effect)

With an energetic distance

$$\Delta E = E_{n,l,m} - E_{n,l,m-1} = \mu_B B$$

between adjacent levels which is determined by the product of Bohr Magneton μ_B and magnetic field strength $\mathbf{B}$.

$$\mu_B = \frac{e\hbar}{2m_e}$$

m_l	Energy
1	$E_0 + \mu_B B$
0	E_0
-1	$E_0 - \mu_B B$

The splitting of the $2l+1$ degenerate m components in an external magnetic field B due to the orbital magnetic moment related to the angular momentum $|l| = \sqrt{l(l+1)}\hbar$ is called **normal Zeeman effect**.

$$\mu_B = \frac{e\hbar}{2m_e}$$

$$E_1 = E_{coul}(n) - \mu_B B$$

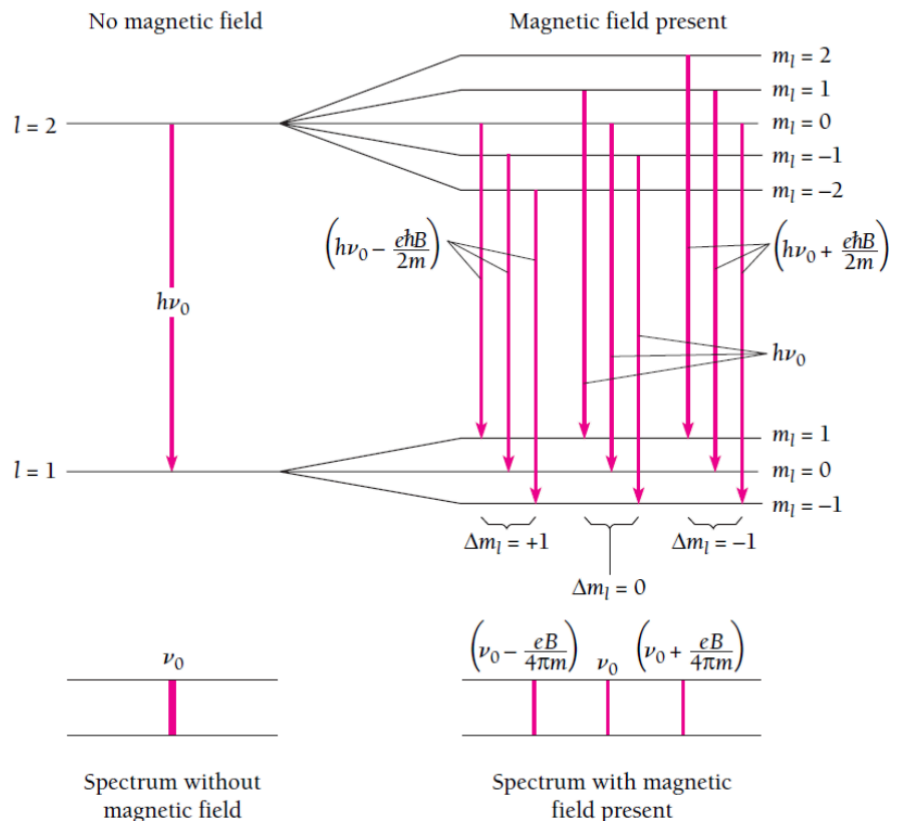
$$v_1 = v_{coul}(n) - \frac{\mu_B B}{h}$$

$$= v_{coul}(n) - \frac{eh}{4\pi m_e} \frac{B}{h}$$

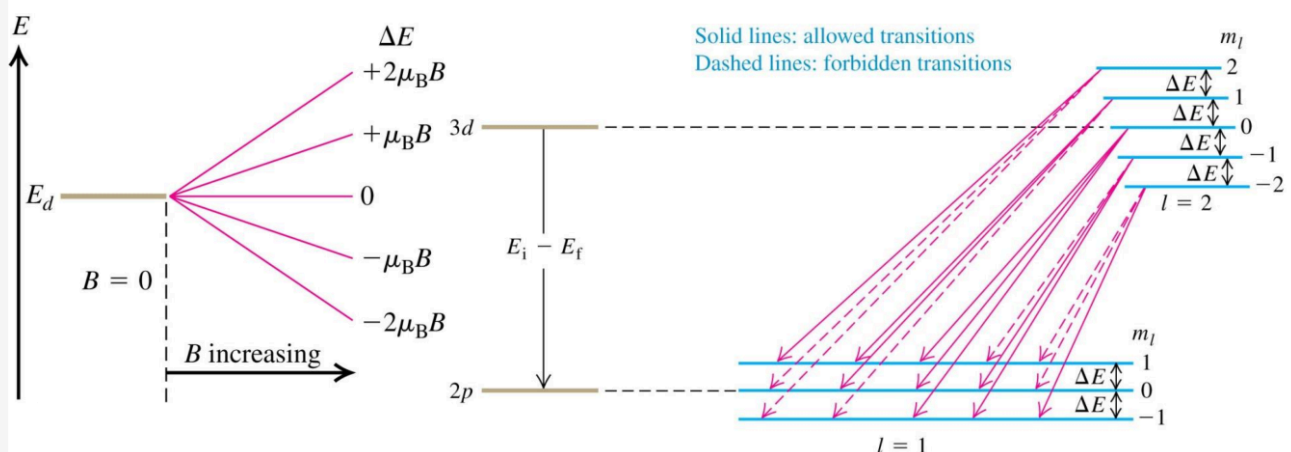
$$= v_{coul}(n) - \frac{eB}{4\pi m_e}$$

$$v_2 = v_{coul}(n)$$

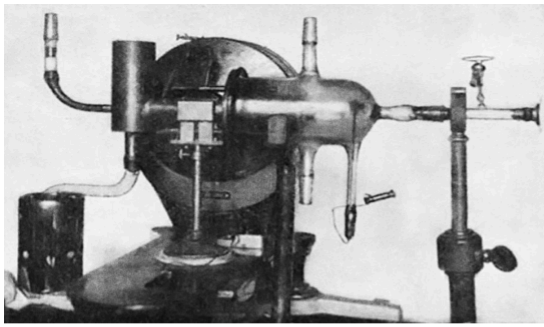
$$v_3 = v_{coul}(n) + \frac{eB}{4\pi m_e}$$



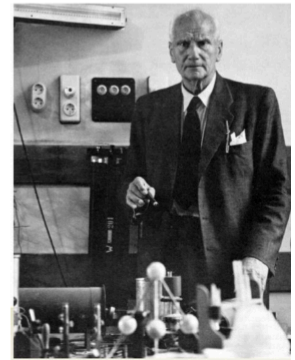
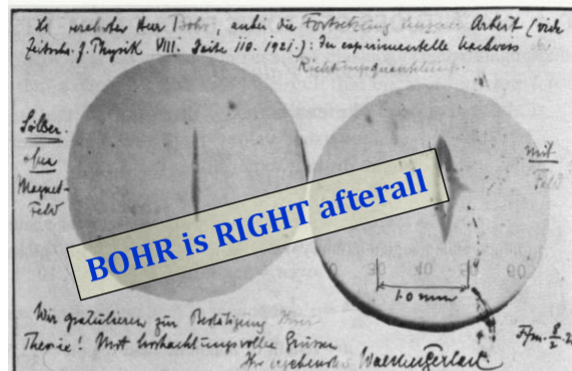
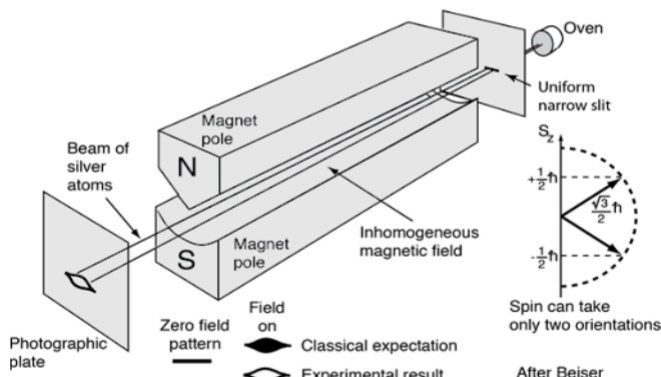
- Lower left figure shows the shift in energy of the five $l=2$ states (each with a different value of m_l) as the magnetic field strength is increased.
- An atom in a magnetic field can make transitions between different states by emitting or absorbing a photon. A transition is *allowed* if l changes by 1 (one) and m_l changes by 0, 1, or -1 . (This is because a photon itself carries angular momentum.)
- A transition is *forbidden* if it violates these *selection rules*.



Stern-Gerlach Experiment



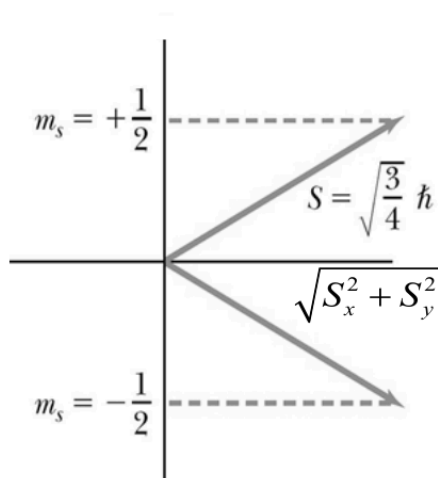
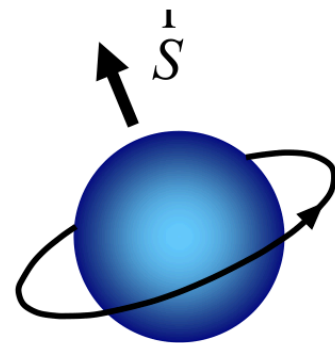
Stern-Gerlach experiment could be explained if assumes that electron has an intrinsic spin and spin must have a value $\frac{1}{2}$.



Intrinsic Spin of Electrons

In analogy with orbital angular momentum, the total spin angular momentum is:

$$|\vec{S}| = \sqrt{s(s+1)} \hbar = \sqrt{3/4} \hbar$$



The z-component of the spinning electron is also analogous to that of the orbiting electron:

So: $S_z = m_s \hbar$

Because the **spin quantum number** m_s has only two values, $\pm\frac{1}{2}$, we say that the electron's spin is either "up" ($m_s = +\frac{1}{2}$) or "down" ($m_s = -\frac{1}{2}$).